

Symmetry and Special Functions

Paul Gustafson

Contents

I	Harmonic Analysis	3
1	Fourier Analysis	3
2	Peter–Weyl Theory	4
3	Symmetric Spaces	5
4	Euclidean Group	7
5	Reductive Groups	8
6	Coadjoint Orbits	9
7	Heisenberg Group	10
8	Unitary Ensembles	12
9	Poincaré Group	14
II	Scattering Theory	15
10	One-Dimensional Scattering	15
11	Potential Scattering	16
12	Coulomb Problem	17
13	Schwarzschild Spacetime	18
III	Hyperbolic Geometry	20
14	$SL_2(\mathbb{R})$	20
15	$SL_2(\mathbb{Z})$	21
IV	Conformal Field Theory	22
16	Virasoro Algebra	22
17	Ising Model	24

V	Quantum Algebra	25
18	Affine Kac–Moody Algebras	25
19	Quantum Groups	27
20	Dunkl Operators	29
21	Double Affine Hecke Algebras	31
VI	Algebraic Geometry	33
22	Algebraic Curves	33
23	Riemann–Roch	34
24	Finite-Gap Spectral Curves	36
VII	Algebraic Topology	37
25	Index Theory	37
26	Cobordism	39
VIII	Tensor Categories	40
27	Geometric Satake	40
28	Fusion Categories	42

Section Map

Section	Geometric symmetry	Special-function decomposition	Physical experiment
I. Harmonic Analysis			
1. Fourier Analysis	LCA translations on G	Pontryagin duality and Fourier kernels	Fraunhofer diffraction
2. Peter–Weyl Theory	$K \times K$ acting on K	Peter–Weyl decomposition and Wigner functions	Rotational spectroscopy
3. Symmetric Spaces	$O(d+1)$ acting on S^d	Jacobi functions and spherical harmonic decomposition	Spherical resonators
4. Euclidean Group	$E(V)$ acting on Euclidean space	Bessel functions and Hankel decomposition	Circular diffraction
5. Reductive Groups	Weyl symmetry of weight space	Highest weights and Weyl characters	Flavor multiplets
6. Coadjoint Orbits	G acting on symplectic orbits	Orbit quantization and coherent-state kernels	Spin precession
7. Heisenberg Group	Central extension of phase-space translations	Stone–von Neumann decomposition and Hermite functions	Oscillator spectroscopy
8. Unitary Ensembles	$U(W)$ acting on Hermitian endomorphisms	Weyl integration and Hermite, sine, and Airy kernels	Quantum-chaotic spectra
9. Poincaré Group	Poincaré symmetry of Minkowski spacetime	Wigner classification and Bessel functions	Correlation-length mass measurement
II. Scattering Theory			
10. One-Dimensional Scattering	Asymptotic translations of the line	Jost decomposition and hypergeometric functions	Transmission and bound-state spectra
11. Potential Scattering	$SO(3)$ acting on $\mathbb{R}^3$	Partial waves and spherical Bessel functions	Hard-sphere scattering
12. Coulomb Problem	$SO(4)$ dynamical symmetry of bound states	Kummer, Laguerre, and Coulomb spectral functions	Hydrogen and Rutherford experiments
13. Schwarzschild Spacetime	$\mathbb{R} \times SO(3)$ acting on the exterior	Regge–Wheeler modes, Heun functions, and parabolic-cylinder functions	Scalar ringdown
III. Hyperbolic Geometry			
14. $SL_2(\mathbb{R})$	$SL_2(\mathbb{R})$ acting on $\mathbb{H}$	Unitary dual, Legendre functions, and the Mehler–Fock transform	Hyperbolic-drum spectra
15. $SL_2(\mathbb{Z})$	Modular symmetry of $\mathbb{H}$	Automorphic spectra, Eisenstein series, and modular forms	Cusp-scattering resonances
IV. Conformal Field Theory			
16. Virasoro Algebra	$\text{Diff}^+(S^1)$ acting on circle fields	Positive-energy modules and Virasoro characters	Conformal finite-size spectra
17. Ising Model	$\mathbb{Z}_2$ acting on spin configurations	Transfer spectra, Toeplitz determinants, and Ising characters	Critical temperature and exponents
V. Quantum Algebra			
18. Affine Kac–Moody Algebras	$LK_c \rtimes S^1$ acting on loop fields	Integrable modules, Weyl–Kac characters, and KZ blocks	WZW finite-size spectra
19. Quantum Groups	$U_v(\mathfrak{sl}_2)$ acting on tensor products	Quantum Peter–Weyl decomposition and basic hypergeometric functions	XXZ spectroscopy
20. Dunkl Operators	Reflection groups acting on Euclidean space	Generalized Bessel and Jack decompositions	Calogero–Sutherland spectroscopy
21. Double Affine Hecke Algebras	DAHA acting on an algebraic torus	Cherednik operators and Macdonald polynomials	Ruijsenaars spectrum
VI. Algebraic Geometry			
22. Algebraic Curves	Elliptic translations on complex tori	Abel–Jacobi inversion and elliptic functions	Nonlinear-pendulum periods
23. Riemann–Roch	$\text{Aut}(X, L)$ acting on sections	Sheaf cohomology, localization, and Bergman kernels	Landau-level density
24. Finite-Gap Spectral Curves	Translations and KdV flows on periodic potentials	Spectral curves, Baker–Akhiezer functions, and Bloch–Floquet theory	Band-gap spectroscopy
VII. Algebraic Topology			
25. Index Theory	Gauge symmetry of connections and coupled operators	Analytic and topological indices with monopole harmonics	Integer Hall conductance
26. Cobordism	Frame symmetry and tangential structures	Pontryagin–Thom theory and eta functions	Invertible-phase response
VIII. Tensor Categories			
27. Geometric Satake	$G[[z]]$ acting on affine-Grassmannian modifications	Satake convolution and Hall–Littlewood polynomials	’t Hooft fusion
28. Fusion Categories	Fusion categories acting on fusion state spaces	Modular sine functions and Verlinde decomposition	Anyon interferometry

Part I

Harmonic Analysis

1 Fourier Analysis

Exercise 1.1 (Hilbert spaces and unitary translations). Work over $\mathbb{C}$. Define normed and Banach spaces, bounded linear maps and their operator norms, inner-product spaces, Cauchy sequences, and completion. Define a Hilbert space to be a complete inner-product space and prove the projection theorem for a closed subspace. Deduce the Riesz representation theorem for bounded linear functionals. For a bounded operator T define its adjoint intrinsically by $\langle Tx, y \rangle = \langle x, T^*y \rangle$ and prove $\|T^*T\| = \|T\|^2$. Define a unitary operator as a surjective linear isometry and a unitary action as a group homomorphism into such operators. Construct $L^2(X, \mu)$ from square-integrable functions modulo equality almost everywhere. If a locally compact group G acts continuously on X preserving μ , prove that pullback defines a unitary action on $L^2(X, \mu)$. Specialize to the additive action of $\mathbb{R}^n$ on itself with Lebesgue measure and prove strong continuity of its translations.

Exercise 1.2 (Locally compact groups and Haar measure). Let G be a locally compact group. Define a positive linear functional on $C_c(G)$ and prove its Riesz–Markov representation by outer approximation. Construct a nonzero positive left-translation-invariant functional by the covering-number argument, apply that representation, and prove the uniqueness needed to obtain left Haar measure dx , unique up to scale. Construct right Haar measure and the continuous homomorphism $\Delta_G : G \rightarrow \mathbb{R}_{>0}$ characterized by

$$\int_G f(xg) dx = \Delta_G(g)^{-1} \int_G f(x) dx.$$

Define convolution and involution by

$$(f * g)(x) = \int_G f(y)g(y^{-1}x) dy, \quad f^*(x) = \Delta_G(x^{-1})\overline{f(x^{-1})},$$

and prove that $L^1(G)$ is a Banach $*$ -algebra and that left translation is unitary on $L^2(G)$. Prove that $\Delta_G = 1$ for abelian and compact groups; in the abelian case adopt additive notation, so that the formulas reduce to additive convolution and reflection involution.

Exercise 1.3 (Pontryagin duality and Fourier decomposition). Define $\widehat{G} = \text{Hom}_{\text{cont}}(G, U(1))$ with the compact-open topology. Normalize Haar measure on $\widehat{G}$ so that

$$\widehat{f}(\chi) = \int_G f(x)\overline{\chi(x)} dx, \quad f(x) = \int_{\widehat{G}} \widehat{f}(\chi)\chi(x) d\chi$$

for the functions obtained by convolving two compactly supported continuous functions. Extend the transform to a unitary $L^2(G) \rightarrow L^2(\widehat{G})$, prove Plancherel and convolution–multiplication, and prove that evaluation identifies G with $\widehat{\widehat{G}}$. Specialize all normalizations to $\mathbb{R}^n$, $\mathbb{Z}^n$, and $\mathbb{T}^n$.

Exercise 1.4 (Poisson summation and Gaussian Fourier kernels). On $\mathbb{R}^n$ use

$$\widehat{f}(\xi) = \int_{\mathbb{R}^n} f(x)e^{-2\pi i x \cdot \xi} dx.$$

Define the Schwartz space by its seminorms, prove Fourier inversion there, and compute $\widehat{e^{-\pi t|x|^2}} = t^{-n/2}e^{-\pi|\xi|^2/t}$. For a lattice L define its covolume and dual lattice intrinsically and prove

$$\sum_{x \in L} f(x) = \frac{1}{\text{covol}(L)} \sum_{\xi \in L^*} \widehat{f}(\xi).$$

Derive the Jacobi transformation $\vartheta(-1/\tau) = (-i\tau)^{1/2}\vartheta(\tau)$ for $\vartheta(\tau) = \sum_{m \in \mathbb{Z}} e^{\pi i m^2 \tau}$, fixing the square-root branch by positivity on the imaginary axis.

Exercise 1.5 (Dirichlet, Fejér, and sinc Fourier kernels). On $\mathbb{T}$ define

$$D_N(\theta) = \sum_{n=-N}^N e^{in\theta} = \frac{\sin((N+1/2)\theta)}{\sin(\theta/2)}, \quad F_N = \frac{1}{N+1} \sum_{j=0}^N D_j.$$

Prove that $F_N \geq 0$, has integral one, and is an approximate identity. Deduce Fejér convergence for continuous functions and the L^2 convergence of Fourier sums. Define $\text{sinc } u = \sin(u)/u$ continuously at zero and compute it as the normalized Fourier transform of an interval.

Exercise 1.6 (Fraunhofer diffraction by rectangular aperture arrays). For monochromatic scalar light of wavelength λ , derive from the Huygens integral in the far-field limit that the amplitude at angle θ is the Fourier transform of the aperture evaluated at θ/λ . Deduce the rectangular-aperture intensity as a product of squared sinc functions. For N equally spaced identical slits, derive the squared Dirichlet-kernel array factor, its principal-peak spacing, and the finite- N peak width. Identify aperture dimensions, slit spacing, wavelength, screen distance, and measured intensity as the calibrated and observed quantities, and state the scalar, paraxial, and far-field approximations.

2 Peter–Weyl Theory

Exercise 2.1 (Finite-dimensional unitary representations of compact groups). Define the tensor product $V \otimes W$ by its universal bilinear map and construct its associativity, unit, and symmetry isomorphisms, writing the last as $\sigma_{V,W} : V \otimes W \rightarrow W \otimes V$. Call V finite-dimensional when there are a vector space $V^\vee$ and evaluation and coevaluation maps

$$\text{ev}_V : V^\vee \otimes V \rightarrow \mathbb{C}, \quad \text{coev}_V : \mathbb{C} \rightarrow V \otimes V^\vee$$

satisfying the two snake identities. Define the trace by

$$\text{Tr}(T) = \text{ev}_V \circ \sigma_{V,V^\vee} \circ (T \otimes 1) \circ \text{coev}_V$$

and put $\dim V = \text{Tr}(1_V)$. Prove cyclicity of trace and its direct-sum and tensor-product formulas. Normalize Haar measure on a compact group K to total mass one. Let $K \times K$ act on K by $(a, b) \cdot g = agb^{-1}$ and prove that normalized Haar measure is invariant. Average an inner product to prove that every finite-dimensional continuous representation is unitarizable and every invariant subspace has an invariant orthogonal complement. Define irreducibility, intertwiners, contragredients, and characters. Prove Schur’s lemma and character orthogonality without choosing coordinates.

Exercise 2.2 (Peter–Weyl decomposition). For an irreducible unitary representation π define its matrix-coefficient map

$$C_\pi : V_\pi \otimes V_\pi^\vee \longrightarrow C(K), \quad C_\pi(v \otimes \phi)(g) = \phi(\pi(g^{-1})v).$$

Prove Schur orthogonality and that $\sqrt{\dim V_\pi} C_\pi$ is an isometry. Use convolution, finite-rank convolution projections, and Stone–Weierstrass to prove the unitary decomposition

$$L^2(K) \simeq \widehat{\bigoplus_{\pi \in \widehat{K}} V_\pi \otimes V_\pi^\vee},$$

with left translation on V_π and right translation on $V_\pi^\vee$. Prove the required Stone–Weierstrass density statement from separation of points and closure under conjugation. Derive Fourier inversion and Plancherel for K .

Exercise 2.3 ($SU(2)$, Wigner functions, and characters). Let U be a two-dimensional Hermitian complex vector space with a unit-norm nonzero alternating form $\varepsilon : U \otimes U \rightarrow \mathbb{C}$. Define $SU(2)$ as the unitary automorphisms preserving ε . On the real Euclidean space of traceless Hermitian endomorphisms of U , with $\langle A, B \rangle = \frac{1}{2} \text{Tr}(AB)$, prove that conjugation maps $SU(2)$ onto the connected orthogonal group, denoted $SO(3)$, with kernel $\{\pm 1\}$. Construct the permutation coinvariant quotient $\text{Sym}^n(V)$ of $V^{\otimes n}$. Construct the irreducibles $V_j = \text{Sym}^{2j}(\mathbb{C}^2)$ for $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$ and define the Wigner functions $D_{mn}^j(g)$ as their distinguished weight matrix coefficients. Derive their Euler-angle formula, orthogonality, completeness, and

$$\chi_j(\theta) = \frac{\sin((2j+1)\theta/2)}{\sin(\theta/2)} = U_{2j}(\cos(\theta/2)),$$

thereby defining the Chebyshev polynomials U_n .

Exercise 2.4 (Clebsch–Gordan and Wigner special functions). Prove $V_{j_1} \otimes V_{j_2} \simeq \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} V_j$. Define Clebsch–Gordan coefficients as the normalized components of these canonical isometric inclusions relative to the distinguished weight lines, and define the Wigner $3j$ and $6j$ symbols from them. Derive their orthogonality, the triangle rules, the Racah recoupling identity, and the integral of three Wigner functions. Prove the Wigner–Eckart theorem by identifying tensor operators with intertwiners.

Exercise 2.5 (Rotational spectroscopy). Let a rigid symmetric-top molecule have moment-of-inertia operator with principal values $I_\perp, I_\perp, I_\parallel$. Quantize its rotational Hamiltonian by the $SU(2)$ action and prove

$$E_{JK} = \frac{\hbar^2}{2I_\perp} J(J+1) + \frac{\hbar^2}{2} (I_\parallel^{-1} - I_\perp^{-1}) K^2.$$

Express electric-dipole transition amplitudes through Wigner $3j$ symbols and derive $\Delta J = 0, \pm 1$, the prohibition $J = 0 \rightarrow J = 0$, and the appropriate ΔK rule for dipoles parallel or perpendicular to the symmetry axis. For a linear rotor deduce the line positions $\nu_{J \rightarrow J+1} = \hbar(J+1)/(2\pi I)$ and the relative line strengths from the $3j$ symbols; these are the quantitative spectroscopic predictions.

3 Symmetric Spaces

Exercise 3.1 (Riemannian symmetric spaces and Gelfand pairs). Define smooth manifolds by compatible smooth atlases, smooth maps by their chart representatives, and tangent vectors intrinsically as derivations of germs. Construct tangent bundles and define a Lie group as a group object in smooth manifolds. Define a Riemannian metric, its volume measure, gradient, and divergence, and use $\Delta = -\text{div grad}$ for the nonnegative Laplace–Beltrami operator. Call a connected Riemannian manifold symmetric when every point x admits an isometry fixing x and

acting as -1 on $T_x X$. Prove that a connected compact symmetric space is a homogeneous space G/K for the identity component of its isometry group and a compact stabilizer K .

For a compact group G and closed subgroup K , define the convolution algebra of continuous K -bi-invariant functions. Call (G, K) a Gelfand pair when this algebra is commutative, and prove that this is equivalent to multiplicity-free decomposition of $L^2(G/K)$. Define the orthogonal group $O(V)$ of a Euclidean space V . For $\dim V = d + 1$ with $d \geq 2$, identify its unit sphere with $O(V)/O(V)_x$, prove that its geodesic symmetries are induced by the orthogonal maps fixing x and negating $x^\perp$, and prove that $(O(d + 1), O(d))$ is a Gelfand pair.

Exercise 3.2 (Gamma and generalized hypergeometric functions). For $\Re s > 0$ define

$$\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt, \quad B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt,$$

derive the functional equation and meromorphic continuation of Γ , and prove $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ together with the Euler reflection and Legendre duplication formulas. Put $(a)_0 = 1$ and $(a)_n = a(a+1)\cdots(a+n-1)$. When no lower parameter is a nonpositive integer, define

$${}_pF_q \left(\begin{matrix} a_1, \dots, a_p \\ b_1, \dots, b_q \end{matrix}; z \right) = \sum_{n \geq 0} \frac{(a_1)_n \cdots (a_p)_n}{(b_1)_n \cdots (b_q)_n} \frac{z^n}{n!}.$$

Determine its disk of convergence for $p \leq q$, $p = q + 1$, and $p > q + 1$, with terminating exceptions, and derive its differential equation in terms of $\vartheta = z\partial_z$:

$$\left[\vartheta \prod_{j=1}^q (\vartheta + b_j - 1) - z \prod_{i=1}^p (\vartheta + a_i) \right] y = 0.$$

Derive the confluence limits ${}_2F_1 \rightarrow {}_1F_1 \rightarrow {}_0F_1$ by sending an upper parameter to infinity while rescaling z . Specialize to the Gauss function ${}_2F_1(a, b; c; z)$ and derive its three regular singular points, Euler integral, Gauss value at $z = 1$, Euler and Pfaff transformations, and connection formula between the Frobenius solutions at 0 and 1, first when the exponent differences are nonintegral and then by limits.

Exercise 3.3 (Jacobi and Gegenbauer functions). For $\alpha, \beta > -1$ define the Jacobi polynomials by

$$P_n^{(\alpha, \beta)}(1 - 2z) = \frac{(\alpha + 1)_n}{n!} {}_2F_1(-n, n + \alpha + \beta + 1; \alpha + 1; z).$$

Derive their differential equation, Rodrigues formula, three-term recurrence, orthogonality for $(1-t)^\alpha(1+t)^\beta dt$, squared norms, and completeness. For $\lambda > 0$ define the Gegenbauer functions by

$$C_n^\lambda(t) = \frac{(2\lambda)_n}{(\lambda + 1/2)_n} P_n^{(\lambda-1/2, \lambda-1/2)}(t)$$

and derive their generating function. Recover the Legendre and Chebyshev families as specializations and express each $SU(2)$ Wigner small- d function as an elementary half-angle factor times a Jacobi polynomial, including its normalization.

Exercise 3.4 (Spherical harmonic decomposition). Define the degree- ℓ polynomial functions on V as $\text{Sym}^\ell(V^\vee)$ acting by evaluation. On the unit sphere $S^d \subset V$, define $\mathcal{H}_\ell$ as the restrictions of those annihilated by the Euclidean Laplacian. Prove intrinsically the orthogonal decomposition

$$L^2(S^d) = \widehat{\bigoplus_{\ell \geq 0} \mathcal{H}_\ell}, \quad \Delta_{S^d}|_{\mathcal{H}_\ell} = \ell(\ell + d - 1),$$

and compute, with a binomial coefficient understood to vanish when its lower argument is negative,

$$N_{d,\ell} = \dim \mathcal{H}_\ell = \binom{\ell + d}{\ell} - \binom{\ell + d - 2}{\ell - 2}.$$

Using the Gelfand-pair decomposition, prove that the orthogonal projection onto $\mathcal{H}_\ell$ has kernel

$$\Pi_\ell(x, y) = \frac{N_{d,\ell}}{\text{vol}(S^d)} \frac{C_\ell^{(d-1)/2}(x \cdot y)}{C_\ell^{(d-1)/2}(1)}.$$

Derive the Funk–Hecke formula and its inversion for zonal kernels. For $d = 2$, define the distinguished spherical harmonics $Y_{\ell m}$ as the right- $SO(2)$ types among the $SO(3)$ matrix coefficients and recover the Legendre addition formula without choosing an auxiliary family.

Exercise 3.5 (Spherical resonator spectroscopy). Let a scalar wave of propagation speed v be confined to a thin spherical resonator of radius R , with dynamics $\partial_t^2 u + (v^2/R^2)\Delta_{S^d} u = 0$. Use the spherical harmonic decomposition and Gegenbauer projection kernels to prove that its angular normal modes have frequencies and degeneracies

$$\omega_\ell = \frac{v}{R} \sqrt{\ell(\ell + d - 1)}, \quad \text{mult}(\omega_\ell) = N_{d,\ell}.$$

For a rotationally symmetric source, express the measured two-point response as a Gegenbauer series and derive its allowed frequencies and angular nodal sets. Separate the scalar thin-shell model, uniform propagation speed, and perfect rotational symmetry from the measured resonance frequencies, degeneracies, and angular correlations.

4 Euclidean Group

Exercise 4.1 (Euclidean motion symmetry and momentum orbits). For a finite-dimensional real inner-product space V , define $E(V) = V \rtimes O(V)$ and its isometric action on the affine Euclidean space modeled on V , and write $d = \dim V$. Show that Fourier transformation intertwines translations with multiplication by characters and rotations with their action on V^* . Prove that the nonzero $O(V)$ -orbits in V^* are the momentum spheres and disintegrate Plancherel measure into radial measure and their invariant probability measures. Apply the compact-group decomposition of each sphere to obtain the orthogonal angular-momentum decomposition of $L^2(V)$. For a transitive locally compact group action on G/H with invariant measure and a unitary H -representation, define the induced representation as the square-integrable equivariant sections of the associated Hilbert bundle and prove that left translation is unitary. Construct the irreducible positive-momentum representations of $E(V)$ by inducing the corresponding translation character tensored with an irreducible representation of its orthogonal stabilizer, and identify the scalar sector with its $O(V)$ -fixed induced functions.

Exercise 4.2 (Bessel functions and Hankel decomposition). For $c \notin \mathbb{Z}_{\leq 0}$ specialize the generalized hypergeometric series to

$${}_0F_1(; c; z) = \sum_{k \geq 0} \frac{z^k}{(c)_k k!}, \quad J_\nu(z) = \frac{(z/2)^\nu}{\Gamma(\nu + 1)} {}_0F_1(; \nu + 1; -z^2/4).$$

Verify directly the double-confluence limit from the Gauss series. Derive Bessel's differential equation, its recurrence and Wronskian relations, and the angular integral that realizes $J_{d/2-1}$ as the spherical

function of the Gelfand pair $(E(d), O(d))$. Continue J_ν in its order and argument, and for $\nu \notin \mathbb{Z}$ define

$$Y_\nu(z) = \frac{\cos(\pi\nu)J_\nu(z) - J_{-\nu}(z)}{\sin(\pi\nu)}, \quad H_\nu^{(1)} = J_\nu + iY_\nu,$$

with integral-order values obtained by continuation. Derive their Wronskian and incoming/outgoing asymptotics.

Use polar disintegration and the spherical harmonic projections to derive the radial Fourier formula

$$\hat{f}(\rho) = 2\pi\rho^{1-d/2} \int_0^\infty f(r) J_{d/2-1}(2\pi r\rho) r^{d/2} dr$$

for radial Schwartz functions, and the order $\ell + d/2 - 1$ Hankel transform on the ℓ th spherical summand. Prove its inversion, Plancherel formula, and completeness kernel from Fourier inversion.

Exercise 4.3 (Circular-aperture diffraction). For a circular aperture of diameter D illuminated by monochromatic scalar light of wavelength λ , use Euclidean rotational symmetry and the order-zero Hankel transform to derive

$$\frac{I(\theta)}{I(0)} = \left[\frac{2J_1(\pi D \sin \theta / \lambda)}{\pi D \sin \theta / \lambda} \right]^2.$$

Prove simplicity of the first positive zero of J_1 , bound it by convergent series or Sturm comparison to obtain $j_{1,1} = 3.831705\dots$, and hence predict the first dark ring $\sin \theta = 1.21967\dots \lambda/D$. Derive the encircled-energy profile from the Bessel recurrence. Distinguish the predicted angular intensity and dark-ring radius from aperture calibration, finite screen distance, polarization, and aberration corrections.

5 Reductive Groups

Exercise 5.1 (Lie algebras, enveloping algebras, and Hopf algebras). Define a complex Lie algebra, Lie ideals, representations, and the adjoint representation. Construct the tensor algebra by its universal property and define

$$U(\mathfrak{g}) = T(\mathfrak{g}) / \langle x \otimes y - y \otimes x - [x, y] \rangle$$

by its universal property among associative algebras receiving a Lie map from $\mathfrak{g}$. Filter $U(\mathfrak{g})$ by tensor degree, construct the canonical graded map $\text{Sym}(\mathfrak{g}) \rightarrow \text{gr } U(\mathfrak{g})$, and prove it is an isomorphism by the rewriting filtration and the Jacobi identity. Deduce the Poincaré–Birkhoff–Witt theorem, induction and restriction of modules, and the construction of a Verma module from a triangular decomposition. Define coalgebras, bialgebras, and Hopf algebras by their universal diagrams, and construct the cocommutative Hopf structure on $U(\mathfrak{g})$ for which every $x \in \mathfrak{g}$ is primitive.

Exercise 5.2 (Root data and Weyl groups). Define an affine complex algebraic group by its finitely generated reduced commutative coordinate Hopf algebra, after defining nilpotent elements and reduced algebras. Define connectedness, unipotent groups, the unipotent radical, and reductivity, and construct its Lie algebra from derivations at the counit. For a connected complex reductive group G , define a maximal torus T , its character and cocharacter lattices, roots, coroots, and the Weyl group $W = N_G(T)/T$. Prove that W is finite, construct a W -invariant Euclidean metric on the real weight space $X^*(T) \otimes_{\mathbb{Z}} \mathbb{R}$ by averaging, and prove that every root reflection is the orthogonal reflection in its root hyperplane. Derive the root-space decomposition of $\mathfrak{g}$ and the rank-one $\mathfrak{sl}_2$ subalgebra associated with each root. Choose a positive system, define simple roots, dominant weights, the Weyl length, and ρ , and prove the chamber and Bruhat decompositions.

Exercise 5.3 (Highest-weight decomposition). Define weight spaces without choosing vectors in them. Prove that every finite-dimensional irreducible G -representation has a unique dominant highest weight. For a Borel algebra $\mathfrak{b} = \mathfrak{t} \oplus \mathfrak{n}_+$ define $M_\lambda = U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} \mathbb{C}_\lambda$ and construct the irreducible representation as its unique irreducible quotient. Use the rank-one strings to prove finite dimensionality exactly for dominant integral weights, complete reducibility, and the highest-weight classification.

Exercise 5.4 (Weyl characters and Schur functions). Define exterior powers as alternating tensor quotients and define the determinant of an endomorphism as the scalar by which it acts on the highest nonzero exterior power. Define a partition as a finite weakly decreasing sequence of nonnegative integers. In the group ring of the weight lattice define $A_\mu = \sum_{w \in W} (-1)^{\ell(w)} e^{w\mu}$. Construct the BGG resolution from Verma modules and use its Euler character to prove

$$\text{ch } V_\lambda = \frac{A_{\lambda+\rho}}{A_\rho}, \quad A_\rho = e^\rho \prod_{\alpha > 0} (1 - e^{-\alpha}).$$

For $G = GL_r$ identify these characters with the Schur functions

$$s_\lambda(x) = \frac{\det(x_i^{\lambda_j + r - j})}{\det(x_i^{r - j})},$$

and derive the tableaux expansion, Cauchy identity, and Littlewood–Richardson tensor-product rule.

Exercise 5.5 (Flavor multiplets and the Gell-Mann–Okubo prediction). Use the $SU(3)$ Weyl character $s_{(2,1)}$ to recover the eight weights of the adjoint representation with their multiplicities. Let isospin $SU(2)$ and hypercharge $U(1)$ be the evident subgroup and show that the octet decomposes as two isodoublets, one isotriplet, and one isosinglet. Assume the leading symmetry-breaking mass operator is an isoscalar in $1 \oplus 8$. Apply the Wigner–Eckart theorem and eliminate its three reduced parameters to derive

$$2(M_N + M_\Xi) = 3M_\Lambda + M_\Sigma.$$

Explain which four observed baryon centroid masses test the relation and separate the exact representation-theoretic multiplicities from the first-order symmetry-breaking assumption.

6 Coadjoint Orbits

Exercise 6.1 (Hamiltonian group actions and moment maps). For a smooth manifold, construct its cotangent bundle, differential forms, pullback, wedge product, and exterior derivative, and prove their independence of charts. Identify the Lie algebra of a Lie group with its left-invariant vector fields. Define symplectic manifolds and Hamiltonian vector fields by $\iota_{X_f} \omega = df$, and the Poisson bracket $\{f, g\} = \omega(X_f, X_g)$. For a Lie-group action define an equivariant moment map $\mu : M \rightarrow \mathfrak{g}^*$ by $d\langle \mu, \xi \rangle = \iota_{\xi_M} \omega$. Prove Noether’s theorem, the Poisson relation for moment-map components, and functoriality under products.

Exercise 6.2 (Kirillov–Kostant–Souriau orbits). For $\lambda \in \mathfrak{g}^*$ identify $T_\lambda(G\lambda)$ with $\mathfrak{g}/\mathfrak{g}_\lambda$ and define

$$\omega_\lambda(\text{ad}_\xi^* \lambda, \text{ad}_\eta^* \lambda) = \langle \lambda, [\xi, \eta] \rangle.$$

Prove that this is well defined, closed, nondegenerate, and has moment map the orbit inclusion. Identify the $SU(2)$ orbits with spheres, compute their symplectic areas intrinsically, and derive the Lie–Poisson equations.

Exercise 6.3 (Symplectic reduction and orbit quantization). For a regular value a of a moment map, construct $M//_a G = \mu^{-1}(a)/G_a$ and prove that its two-form is characterized by pullback. Define a prequantum Hermitian line bundle by a unitary connection of curvature $-i\omega/\hbar$. Prove the integrality condition and, for an $SU(2)$ orbit of area $2\pi\hbar n$, realize its quantization as the right-weight- n functions on $SU(2)$ annihilated by the positive-root operator. Use Peter–Weyl to identify this space with $\text{Sym}^n(\mathbb{C}^2)^\vee$. Derive quantization commutes with reduction for tensor products of these rank-one orbits and recover the triangle rule.

Exercise 6.4 (Spin coherent states and their kernels). Let J_1, J_2, J_3 be the infinitesimal $SU(2)$ generators normalized by $[J_a, J_b] = i\hbar\epsilon_{abc}J_c$. For $j = n/2$, define the normalized spin coherent state $|x\rangle$ as the image in $\text{Sym}^n(\mathbb{C}^2)$ of a unit vector projecting to $x \in S^2$. Prove

$$|\langle x | y \rangle|^2 = \left(\frac{1 + x \cdot y}{2} \right)^{2j}, \quad \frac{2j+1}{4\pi} \int_{S^2} |x\rangle \langle x| d\Omega = 1.$$

Expand a rotated coherent state into the J_3 weight lines and show that its probabilities are the binomial kernel

$$p_m(\theta) = \binom{2j}{j+m} \cos^{2(j+m)} \frac{\theta}{2} \sin^{2(j-m)} \frac{\theta}{2}.$$

Define the Krawtchouk polynomials by

$$\sum_{n=0}^N \binom{N}{n} K_n(x; p, N) t^n = \left(1 - \frac{1-p}{p} t \right)^x (1+t)^{N-x}.$$

Identify the associated rotation coefficients with these polynomials at $p = \sin^2(\theta/2)$, identify $K_n(x; p, N) = {}_2F_1(-n, -x; -N; 1/p)$, and derive their difference equation and binomial orthogonality.

Exercise 6.5 (Spin precession and Stern–Gerlach intensities). For these infinitesimal generators let the Zeeman Hamiltonian be $H = -\gamma \mathbf{B} \cdot \mathbf{J}$. Solve the Hamiltonian orbit equation and the quantum evolution to obtain precession frequency $\omega_L = |\gamma| |\mathbf{B}|$. If a beam polarized along x is measured along an axis making angle θ with x , prove that the $2j+1$ separated intensities are precisely the displayed $p_m(\theta)$. Thus the coherent-state/Krawtchouk kernel predicts both the line count and every relative Stern–Gerlach intensity.

7 Heisenberg Group

Exercise 7.1 (Central extensions of phase space). Let (W, ω) be a finite-dimensional real symplectic vector space. Define the Heisenberg group

$$\text{Heis}(W) = W \times \mathbb{R}, \quad (w, s)(w', s') = (w + w', s + s' + \tfrac{1}{2}\omega(w, w')).$$

Prove associativity, identify its center and commutator, and show that its Lie algebra satisfies $[(w, s), (w', s')] = (0, \omega(w, w'))$. Prove that every continuous central extension of the additive group W with this commutator is equivalent to this one after changing its section by a continuous coboundary. Show that the quotient by the center acts transitively on W by symplectic translations, identifying W as the homogeneous classical phase space of this symmetry.

Exercise 7.2 (C^* -algebras and spectral measures). Define a unital C^* -algebra to be a complete normed complex algebra A with an involution satisfying $\|a^*a\| = \|a\|^2$. For commutative A , define invertibility, the spectrum $\text{spec}(a) = \{z \in \mathbb{C} : a - z1 \text{ is not invertible}\}$, its character space $\hat{A}$, and the Gelfand transform. Prove from maximal ideals and the C^* -identity that $A \simeq C(\hat{A})$. Prove that the bounded operators on a Hilbert space form the C^* -algebra $\mathcal{B}(\mathcal{H})$. For a bounded self-adjoint operator, where self-adjoint means $H = H^*$, define $C^*(1, H)$ as the norm closure of the algebra generated by 1 and H . Identify its character space with $\text{spec}(H)$. Define a projection-valued measure by normalization and countable orthogonal additivity in the strong operator topology. Apply the Riesz–Markov theorem to construct the unique such measure P satisfying

$$f(H) = \int_{\text{spec}(H)} f(E) dP(E), \quad H = \int_{\text{spec}(H)} E dP(E).$$

Prove the bounded Borel functional calculus, spectral mapping, and $\langle f(H)x, y \rangle = \int f d\mu_{x,y}$, where $\mu_{x,y}(B) = \langle P(B)x, y \rangle$.

Exercise 7.3 (Self-adjoint operators and one-parameter unitary groups). For a densely defined operator D on a Hilbert space, define its graph, adjoint, closure, resolvent, and the conditions of being symmetric, self-adjoint, and essentially self-adjoint. Prove the closed-graph criterion for the adjoint. For closed symmetric D , construct its Cayley transform from $\text{im}(D + i)$ to $\text{im}(D - i)$ and prove that it extends to a unitary operator exactly when D is self-adjoint. Apply the bounded spectral theorem to the Cayley transform and construct the unique projection-valued measure satisfying

$$D = \int_{\mathbb{R}} \lambda dP(\lambda), \quad \text{Dom}(D) = \left\{ x : \int_{\mathbb{R}} \lambda^2 d\mu_x(\lambda) < \infty \right\}.$$

Develop the measurable functional calculus, including its domain rule and spectral mapping. Prove Stone's theorem: every strongly continuous unitary group has a unique self-adjoint generator D with $U(t) = e^{-itD}$, and conversely. Identify its spectral resolution with decomposition of time translation into the characters $t \mapsto e^{-it\lambda}$.

Exercise 7.4 (Weyl relations and the Schrödinger representation). For a real finite-dimensional vector space Q , put $W = Q \oplus Q^*$ and $\omega((q, p), (q', p')) = p(q') - p'(q)$. On $L^2(Q)$ with Lebesgue measure fixed by a density, define intrinsically

$$(T_q f)(x) = f(x - q), \quad (M_p f)(x) = e^{ip(x)/\hbar} f(x),$$

and $W(q, p) = e^{-ip(q)/(2\hbar)} M_p T_q$. Prove

$$W(w)W(w') = e^{i\omega(w, w')/(2\hbar)} W(w + w').$$

On the Schwartz core define $Q_p f(x) = p(x)f(x)$ and $P_q f = -i\hbar \frac{d}{dt}|_{t=0} f(x + tq)$. Differentiate the Weyl relations to obtain $[P_q, Q_p] = -i\hbar p(q)1$. For a unit vector x in a common invariant domain of self-adjoint A, B , put $\langle A \rangle_x = \langle Ax, x \rangle$ and prove the Robertson uncertainty inequality directly from positivity of $\|(A - \langle A \rangle_x)x + it(B - \langle B \rangle_x)x\|^2$.

Exercise 7.5 (Stone–von Neumann decomposition). Call a unitary representation of $\text{Heis}(W)$ regular if its one-parameter subgroups are strongly continuous and assume its center acts by $e^{is/\hbar}$. For a Lagrangian subspace $Q^* \subset W$, construct the joint projection-valued measure of its commuting translations from the one-parameter spectral theorem. Prove the imprimitivity relation for the complementary translations, identify its spectral multiplicity as constant, and construct a unitary intertwiner with a Hilbert direct sum of Schrödinger representations. Deduce that every irreducible regular representation with this central character is unitarily equivalent to the Schrödinger representation.

Exercise 7.6 (Green functions and spectral measures). Define complex differentiability and holomorphic and meromorphic Banach-valued functions. For $z \notin \text{spec}(D)$ define the resolvent and scalar Green function

$$R_D(z) = (D - z)^{-1}, \quad G_x(z) = \langle R_D(z)x, x \rangle = \int_{\mathbb{R}} \frac{d\mu_x(E)}{E - z}.$$

Prove the resolvent identity, holomorphy, and $\Im G_x(z) > 0$ for $\Im z > 0$. Prove the Stieltjes inversion formula

$$\mu_x((a, b)) = \lim_{\epsilon \downarrow 0} \frac{1}{\pi} \int_a^b \Im G_x(E + i\epsilon) dE$$

when the endpoints are not atoms, and prove the converse Herglotz representation under the decay $G(iy) = O(y^{-1})$.

Exercise 7.7 (Hermite functions and the Mehler kernel). For $Q = \mathbb{R}$, take the closures of $Xf = xf$ and $Pf = -i\hbar f'$ from the Schwartz core. For

$$H = \frac{P^2}{2m} + \frac{m\omega^2 X^2}{2}, \quad a = \frac{m\omega X + iP}{\sqrt{2m\hbar\omega}},$$

prove $[a, a^*] = 1$ and $H = \hbar\omega(a^*a + 1/2)$. Define the Hermite polynomials by $H_n(u) = (-1)^n e^{u^2} (d/du)^n e^{-u^2}$ and the normalized Hermite functions h_n . Derive their differential equation, recurrence, orthogonality, Mehler kernel, and completeness without calling them a basis. Identify the even and odd Hermite polynomials with ${}_1F_1(-n; 1/2; u^2)$ and $u {}_1F_1(-n; 3/2; u^2)$, including their constants.

Exercise 7.8 (Oscillator spectroscopy). Use the Hermite decomposition to prove $E_n = \hbar\omega(n + 1/2)$. Couple the oscillator to a weak electric field by $V(t) = -qE_0 X \cos(\Omega t)$ and derive from the ladder relations that only $n \leftrightarrow n \pm 1$ transitions occur, with

$$|\langle n+1 | X | n \rangle|^2 = \frac{\hbar}{2m\omega}(n+1).$$

For a state prepared in level n , define its positive-frequency dipole spectral measure and Green function by

$$d\mu_n(\nu) = |\langle n+1 | X | n \rangle|^2 \delta_\omega(d\nu), \quad G_n(z) = \int_{\mathbb{R}} \frac{d\mu_n(\nu)}{\nu - z}.$$

For frequency resolution $\gamma > 0$, prove that $\pi^{-1} \Im G_n(\Omega + i\gamma)$ is a Lorentzian centered at $\Omega = \omega$, with full width at half maximum 2γ and integrated intensity $\hbar(n+1)/(2m\omega)$. Average these profiles with the Gibbs weights $p_n = Z^{-1} e^{-\beta E_n}$ to obtain the thermal response. The line position, state-resolved intensity ratios, and thermal intensity are the spectroscopic predictions.

8 Unitary Ensembles

Exercise 8.1 (Unitary symmetry and spectral measures). On the real inner-product space of Hermitian endomorphisms of a finite-dimensional Hermitian space W , let $U(W)$ act by conjugation. For a real confining function $\mathcal{V}$, prove that

$$d\mathbb{P}(H) = Z^{-1} e^{-\text{Tr}(\mathcal{V}(H))} dH$$

is invariant whenever it is integrable. Disintegrate this measure along conjugacy orbits and identify the unordered eigenvalue map without choosing eigenvectors.

Exercise 8.2 (Weyl integration and eigenvalue repulsion). Compute the normal Jacobian of the conjugation orbit through a simple-spectrum Hermitian operator and prove the Weyl integration formula

$$\int F(H) dH = C_n \int_{\mathbb{R}^n} F(\lambda) \prod_{i < j} (\lambda_i - \lambda_j)^2 \prod_i d\lambda_i$$

for conjugation-invariant F . Determine C_n from a Gaussian integral. Show that the squared Vandermonde forces quadratic repulsion at collisions and rewrite the eigenvalue density as a determinant using monic polynomials orthogonal for $e^{-\mathcal{V}(x)} dx$.

Exercise 8.3 (Orthogonal-polynomial and Christoffel–Darboux kernels). Let p_k be the monic orthogonal polynomials for $e^{-\mathcal{V}(x)} dx$, assumed positive with all moments, and let $h_k = \|p_k\|^2$. Derive the three-term recurrence and prove the Christoffel–Darboux identity

$$\sum_{k=0}^{n-1} \frac{p_k(x)p_k(y)}{h_k} = \frac{p_n(x)p_{n-1}(y) - p_{n-1}(x)p_n(y)}{h_{n-1}(x-y)}.$$

For $\phi_k = p_k e^{-\mathcal{V}/2} / \sqrt{h_k}$ define $K_n(x, y) = \sum_{k < n} \phi_k(x)\phi_k(y)$. Apply the Andreief determinant identity to prove that every r -point eigenvalue correlation is $\det(K_n(x_i, x_j))_{i,j \leq r}$ and that K_n is an orthogonal-projection kernel.

Exercise 8.4 (Hermite, sine, and Airy kernels). For holomorphic and meromorphic functions, define piecewise smooth contour integrals, winding numbers, and residues. Prove Goursat’s theorem, the Cauchy integral formula, local power-series expansion, the residue theorem, and uniqueness of analytic continuation along a path. Derive contour deformation through regions of holomorphy. For $\mathcal{V}(x) = x^2$, identify p_k with the Hermite polynomials and compute their normalization. Use steepest descent in their contour representation to derive the bulk and soft-edge limits

$$\frac{1}{\rho_n(E)} K_n \left(E + \frac{u}{\rho_n(E)}, E + \frac{v}{\rho_n(E)} \right) \longrightarrow \frac{\sin \pi(u-v)}{\pi(u-v)},$$

$$\frac{1}{c_n} K_n \left(a_n + \frac{u}{c_n}, a_n + \frac{v}{c_n} \right) \longrightarrow K_{\text{Ai}}(u, v) = \frac{\text{Ai}(u) \text{Ai}'(v) - \text{Ai}'(u) \text{Ai}(v)}{u-v}.$$

Define $\text{Ai}(u) = (2\pi i)^{-1} \int_{\mathcal{C}} e^{t^3/3 - ut} dt$ on the standard Airy contour and derive both real asymptotic regimes and its differential equation.

Exercise 8.5 (Fredholm determinants and quantum-chaotic spectra). Define the singular values of a compact operator from the spectral theorem for $|A| = (A^*A)^{1/2}$ and call A trace class when their sum is finite. For positive trace-class B , define $\text{Tr}(B) = \sum_{\lambda > 0} \lambda \text{rank } P_{\{\lambda\}}$ and extend the trace to every trace-class operator through its decomposition into the real and imaginary parts and their positive and negative parts. Prove that the result is independent of the decomposition, that $|\text{Tr}(A)| \leq \text{Tr}|A|$, and that $\text{Tr}(AB) = \text{Tr}(BA)$ whenever one factor is trace class and the other bounded. Construct the Hilbert exterior powers $\wedge^r \mathcal{H}$ as the ranges of the antisymmetrizing projections on Hilbert tensor powers and show that $\wedge^r A$ is trace class. With no positivity or self-adjointness assumption, define

$$\det(1 + A) = \sum_{r \geq 0} \text{Tr}(\wedge^r A)$$

and prove absolute convergence, multiplicativity, and trace-norm continuity. Define a point process as a random locally finite counting measure and its correlation measures by factorial moments.

Call it determinantal with kernel K when its correlation densities are $\det(K(x_i, x_j))_{i,j \leq r}$. Derive its generating functional and prove the gap formula $\mathbb{P}(N_D = 0) = \det(1 - 1_D K 1_D)$. In the sine-kernel limit derive $R_2(s) = 1 - (\sin \pi s / (\pi s))^2 = \pi^2 s^2 / 3 + O(s^4)$; at the edge define $F_2(s) = \det(1 - K_{\text{Ai}})_{L^2(s, \infty)}$. Define the Hastings–McLeod solution by

$$q''(s) = sq(s) + 2q(s)^3, \quad q(s) \sim \text{Ai}(s) \quad (s \rightarrow +\infty),$$

prove existence and uniqueness from its Volterra equation and resolvent estimates, and derive $F_2(s) = \exp(-\int_s^\infty (x-s)q(x)^2 dx)$. For a quantum system whose unfolded spectrum has no antiunitary symmetry, formulate the predictions of quadratic level repulsion, the sine-kernel two-point curve, and the F_2 largest-level law, separating the random-matrix modeling hypothesis from the measured spacings.

9 Poincaré Group

Exercise 9.1 (Minkowski isometries and their Casimirs). Let Minkowski spacetime be an oriented, time-oriented affine space modeled on (M, g) with signature $(-, +, +, +)$. Define $\mathcal{P}^+ = M \rtimes SO^+(M, g)$, derive its Lie algebra, and construct its metric volume form and Hodge star, then construct its two Casimirs from momentum P and the Pauli–Lubanski covector $W = *(P^\flat \wedge J)$, fixing this normalization. Prove $P^2 = -m^2 c^2$ and $W^2 = -m^2 c^2 \hbar^2 j(j+1)$ on a massive spin- j irreducible representation.

Exercise 9.2 (Wigner classification by little groups). Prove the imprimitivity theorem for an induced unitary representation by disintegrating the translation spectral measure. Apply it to the mass shells in M^* : identify the massive little group with $SU(2)$ after passing to the double cover, the massless little group with $\text{Spin}(2) \rtimes \mathbb{R}^2$, where $\text{Spin}(2) = U(1) \rightarrow SO(2)$ is the angle-doubling cover, and derive the massive spin and massless helicity families. State explicitly the exclusion of continuous-spin representations when the translational part of the massless little group acts trivially.

Exercise 9.3 (Klein–Gordon modified Bessel functions). Put $\mu = mc/\hbar$ and $\square = -c^{-2}\partial_t^2 + \Delta$. Derive $(\square - \mu^2)\phi = 0$ from the massive Casimir. For $\Re z > 0$ define the modified Bessel function

$$K_\nu(z) = \int_0^\infty e^{-z \cosh u} \cosh(\nu u) du$$

for $\Re z > 0$. Derive its differential equation, recurrence, analytic continuation, and connection with the outgoing Hankel function. Use the Fourier decomposition of the positive mass shell to derive, at equal time and spatial separation $r > 0$,

$$\Delta_+(0, r) = \frac{mc}{4\pi^2 \hbar r} K_1(\mu r).$$

Derive the small- and large-argument asymptotics needed below.

Exercise 9.4 (Relativistic correlation length). For a free massive scalar field, define the vacuum two-point observable by the positive-energy mass-shell integral and derive the preceding K_1 formula without choosing an inertial frame until evaluating the invariant spacelike distance. Prove

$$\Delta_+(x, y) \sim \frac{\sqrt{m}}{4\sqrt{2}\pi^{3/2}(\hbar/c)^{1/2}r^{3/2}} e^{-r/\lambda_C}, \quad \lambda_C = \frac{\hbar}{mc},$$

up to the normalization fixed in the previous exercise. Show that the slope of $\log(r^{3/2}\Delta_+)$ against invariant separation is $-1/\lambda_C$; this is a quantitative mass determination from spacelike correlation data.

Part II

Scattering Theory

10 One-Dimensional Scattering

Exercise 10.1 (Asymptotic translation symmetry and scattering channels). Let $\mathbb{R}$ act on its configuration line by translations and on $L^2(\mathbb{R})$ by pullback. Let

$$H_0 = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2}, \quad H = H_0 + V,$$

where V is real and $\int_{\mathbb{R}} (1 + |x|) |V(x)| dx < \infty$. Construct H from its closed semibounded quadratic form after defining densely defined, closed, and semibounded forms and proving their self-adjoint representation theorem. For a semibounded symmetric operator, close its form and construct its Friedrichs extension. Prove that H_0 is translation-invariant and that V breaks this symmetry only by a short-range interaction, and prove that the absolutely continuous spectrum on $(0, \infty)$ has two asymptotic translation channels. Construct the wave operators

$$\Omega_{\pm} = \text{s-lim}_{t \rightarrow \pm\infty} e^{iHt/\hbar} e^{-iH_0t/\hbar}$$

first on Fourier transforms of compactly supported functions away from zero, using the Duhamel formula and the one-dimensional dispersive estimate, and then by density. Prove isometry, intertwining, and completeness from the Volterra resolvent. Define $S = \Omega_+^* \Omega_-$ and prove that it commutes with H_0 . Show that time reversal makes the two-channel scattering matrix symmetric and that an even potential further decomposes it into parity channels.

Exercise 10.2 (Jost solutions and the scattering matrix). For $k > 0$ define the Jost solutions as the unique solutions of

$$-\frac{\hbar^2}{2m} f'' + Vf = \frac{\hbar^2 k^2}{2m} f, \quad f_{\pm}(k, x) e^{\mp i k x} \rightarrow 1 \quad (x \rightarrow \pm\infty),$$

and construct them from their Volterra integral equations. Use their Wronskians to define transmission and left and right reflection amplitudes, fixing the convention

$$f_+(k, x) = t(k)^{-1} e^{i k x} + t(k)^{-1} r_-(k) e^{-i k x} + o(1) \quad (x \rightarrow -\infty).$$

Prove that these coefficients form the fiber of S , derive unitarity and reciprocity, and meromorphically continue them to $\Im k > 0$. Prove that poles of $t(k)$ at $k = i\kappa$ are bound states of energy $-\hbar^2 \kappa^2 / (2m)$, derive their norming constants from the residues, and identify lower-half-plane continuation poles as resonances.

Exercise 10.3 (Pöschl–Teller hypergeometric functions). For

$$V(x) = -\frac{\hbar^2}{2ma^2} \lambda(\lambda + 1) \operatorname{sech}^2(x/a), \quad \lambda > 0,$$

put $z = (1 - \tanh(x/a))/2$ and reduce the Jost equation to the Gauss equation. Identify its Gauss hypergeometric parameters and specialize the Euler integral and the connection formula between $z = 0$ and $z = 1$. Express both Jost solutions through these functions and obtain, with $p = ka$,

$$t(p) = \frac{\Gamma(1 + \lambda - ip) \Gamma(-\lambda - ip)}{\Gamma(1 - ip) \Gamma(-ip)},$$

in the preceding asymptotic convention. Use the gamma reflection formula to prove

$$|t(p)|^2 = \frac{\sinh^2(\pi p)}{\sinh^2(\pi p) + \sin^2(\pi \lambda)}, \quad |r(p)|^2 = \frac{\sin^2(\pi \lambda)}{\sinh^2(\pi p) + \sin^2(\pi \lambda)}.$$

Exercise 10.4 (Reflectionless transmission spectroscopy). For integral $\lambda = N$, prove that the Pöschl–Teller potential is reflectionless at every positive energy and that its bound-state poles give

$$E_n = -\frac{\hbar^2}{2ma^2}(N-n)^2, \quad n = 0, \dots, N-1.$$

For nonintegral λ , locate the bound-state poles with $n \in \mathbb{Z}_{\geq 0}$ and $n < \lambda$ and the continued resonance poles, and derive the threshold and high-energy asymptotics of the reflection probability. Formulate a one-dimensional matter-wave or microwave-guide experiment that measures transmitted and reflected flux as a function of k : the model predicts the displayed transmission curve, exact reflectionlessness at integral N , and the bound-state line positions. Separate calibration of a and the potential depth, finite preparation width, absorption, and truncation of the ideal sech^2 profile from these predicted observables.

11 Potential Scattering

Exercise 11.1 ($SO(3)$ symmetry and radial operators). On $L^2(\mathbb{R}^3)$ let $SO(3)$ act by pullback. For a central potential $V(r)$, prove that the Schrödinger operator commutes with this action. Use the spherical-harmonic decomposition to obtain the intrinsic orthogonal sum over angular momentum and the radial operators

$$H_\ell = -\frac{\hbar^2}{2m} \frac{d^2}{dr^2} + \frac{\hbar^2 \ell(\ell+1)}{2mr^2} + V(r)$$

on $L^2(\mathbb{R}_+, dr)$ with the regular boundary condition.

Exercise 11.2 (Partial-wave decomposition and phase shifts). For a short-range real potential, construct the regular radial solution by a Volterra equation and prove that for $k > 0$ it has the asymptotic form

$$u_\ell(k, r) = A_\ell(k) \sin(kr - \ell\pi/2 + \delta_\ell(k)) + o(1).$$

Define $S_\ell(k) = e^{2i\delta_\ell(k)}$, fix $\delta_\ell = 0$ for the free operator, and use the conserved Wronskian to prove $|S_\ell| = 1$. Derive

$$f(\theta) = \frac{1}{2ik} \sum_{\ell \geq 0} (2\ell+1)(S_\ell - 1) P_\ell(\cos \theta), \quad \sigma_{\text{tot}} = \frac{4\pi}{k^2} \sum_{\ell \geq 0} (2\ell+1) \sin^2 \delta_\ell.$$

Exercise 11.3 (Spherical Bessel functions and hard-sphere phase shifts). Define

$$j_\ell(z) = \sqrt{\frac{\pi}{2z}} J_{\ell+1/2}(z), \quad y_\ell(z) = \sqrt{\frac{\pi}{2z}} Y_{\ell+1/2}(z),$$

and derive their radial equation, recurrences, Wronskian, small-argument behavior, and incoming/outgoing combinations. For an impenetrable sphere of radius a , impose the exterior boundary condition $u_\ell(a) = 0$ and prove

$$\tan \delta_\ell(k) = \frac{j_\ell(ka)}{y_\ell(ka)}.$$

Use the Bessel asymptotics to determine the low- and high-energy behavior of these phase shifts and verify the optical theorem from the partial-wave decomposition.

Exercise 11.4 (Hard-sphere scattering). Use the preceding Bessel phase shifts to derive the angular amplitude

$$f_a(\theta) = \frac{1}{k} \sum_{\ell \geq 0} (2\ell + 1) e^{i\delta_\ell} \sin \delta_\ell P_\ell(\cos \theta)$$

and the total cross section

$$\sigma_{\text{tot}}(k) = \frac{4\pi}{k^2} \sum_{\ell \geq 0} (2\ell + 1) \frac{j_\ell(ka)^2}{j_\ell(ka)^2 + y_\ell(ka)^2}.$$

Prove $\sigma_{\text{tot}}(k) \rightarrow 4\pi a^2$ as $ka \downarrow 0$, determine the first corrections and the diffraction oscillations at finite ka , and formulate a matter-wave or acoustic scattering experiment measuring the differential and total cross sections. Separate the perfectly reflecting spherical-boundary model and incident-wave calibration from the measured angular intensities and cross section.

12 Coulomb Problem

Exercise 12.1 (Runge–Lenz symmetry of the Coulomb Hamiltonian). Let V be an oriented three-dimensional Euclidean space, let X and P be the position and momentum operators in the Schrödinger representation, and use the metric and orientation to define their cross product. Construct the attractive Coulomb Hamiltonian by its closed semibounded form,

$$H = \frac{P^2}{2m} - \frac{\kappa}{|X|}, \quad \kappa > 0,$$

and on a common invariant core define

$$L = X \times P, \quad A = \frac{1}{2m}(P \times L - L \times P) - \kappa \frac{X}{|X|}.$$

Prove the Hardy inequality needed to make the Coulomb term form-bounded with relative bound zero and hence justify the stated self-adjoint realization. For $a, b \in V$, derive

$$[a \cdot L, b \cdot L] = i\hbar(a \times b) \cdot L, \quad [a \cdot L, b \cdot A] = i\hbar(a \times b) \cdot A, \quad [a \cdot A, b \cdot A] = -\frac{2i\hbar H}{m}(a \times b) \cdot L,$$

together with $L \cdot A = 0$ and $A^2 = \kappa^2 + 2H(L^2 + \hbar^2)/m$. On the negative spectral subspace rescale A and prove that L and the rescaled Runge–Lenz operator generate $\mathfrak{so}(4) \simeq \mathfrak{su}(2) \oplus \mathfrak{su}(2)$. Use its two Casimirs to derive

$$E_n = -\frac{m\kappa^2}{2\hbar^2 n^2}, \quad \text{mult}(E_n) = n^2, \quad n \geq 1.$$

Exercise 12.2 (Kummer and Laguerre spectral functions). Specialize the generalized hypergeometric series and verify its Gauss confluence formula

$${}_1F_1(a; c; z) = \lim_{b \rightarrow \infty} {}_2F_1(a, b; c; z/b) = \sum_{j \geq 0} \frac{(a)_j}{(c)_j j!} z^j.$$

Derive Kummer’s equation, its Euler integral and connection formula, and the confluence of the Gauss singular points, initially under $\Re c > \Re a > 0$ for the integral and then by meromorphic continuation. For $\alpha > -1$ define

$$L_j^\alpha(z) = \frac{(\alpha + 1)_j}{j!} {}_1F_1(-j; \alpha + 1; z),$$

and derive its Rodrigues formula, recurrence, differential equation, orthogonality for $z^\alpha e^{-z} dz$, squared norm, and completeness.

Put $a_0 = \hbar^2/(m\kappa)$. Separate the Coulomb Hamiltonian by the spherical decomposition and prove that its normalized bound radial functions are

$$R_{n\ell}(r) = N_{n\ell} e^{-r/(na_0)} \left(\frac{2r}{na_0} \right)^\ell L_{n-\ell-1}^{2\ell+1} \left(\frac{2r}{na_0} \right), \quad 0 \leq \ell < n.$$

Determine $N_{n\ell}$, prove orthogonality and completeness on the bound spectral subspace, and identify the resulting $SO(4)$ decomposition with the tensor product of the two spin- $(n-1)/2$ representations obtained from the Runge–Lenz symmetry.

Exercise 12.3 (Coulomb scattering functions). For the repulsive potential $V(r) = \kappa/r$, set $\rho = kr$ and $\eta = m\kappa/(\hbar^2 k)$. Define the regular Coulomb function by

$$F_\ell(\eta, \rho) = C_\ell(\eta) \rho^{\ell+1} e^{-i\rho} {}_1F_1(\ell+1-i\eta; 2\ell+2; 2i\rho),$$

choosing $C_\ell(\eta) > 0$ so that its large- ρ sine amplitude is one. Derive the radial equation and specialize the Kummer connection formula to obtain $\sigma_\ell = \arg \Gamma(\ell+1+i\eta)$ and $\sigma_\ell - \sigma_{\ell-1} = \arctan(\eta/\ell)$. Sum the resulting Legendre series by its generating function and prove that its long-range phase is compatible with flux conservation.

Exercise 12.4 (Hydrogen spectroscopy and Rutherford scattering). For the attractive potential with $\kappa = Ze^2/(4\pi\epsilon_0)$, use the Laguerre spectral functions and the Wigner–Eckart theorem to derive the electric-dipole rules $\Delta\ell = \pm 1$ and $\Delta m = 0, \pm 1$, and express every radial line strength by a terminating Laguerre integral. Prove that the line frequencies are

$$\nu_{n \rightarrow n'} = \frac{m\kappa^2}{4\pi\hbar^3} \left| \frac{1}{n^2} - \frac{1}{n'^2} \right|,$$

with the reduced mass replacing m for a moving nucleus.

For the repulsive potential, use the Coulomb connection coefficient to derive the amplitude, up to an unobservable constant phase,

$$f_C(\theta) = -\frac{\eta}{2k \sin^2(\theta/2)} \exp \left[-2i\eta \log \sin \frac{\theta}{2} + 2i\sigma_0 \right],$$

and hence

$$\frac{d\sigma}{d\Omega} = \left(\frac{Z_1 Z_2 e^2}{16\pi\epsilon_0 E} \right)^2 \csc^4 \frac{\theta}{2}.$$

Identify spectroscopic frequencies and intensities, beam energy, scattering angle, and detector solid angle as measured quantities. State separately the nonrelativistic, point-nucleus, spin-free, and radiative limitations of the two predictions.

13 Schwarzschild Spacetime

Exercise 13.1 (Lorentzian geometry and spherical vacuum metrics). Define a Lorentzian metric, its Levi–Civita connection by the Koszul identity, curvature, Ricci tensor, and geodesics. Derive the

Jacobi equation from a variation of geodesics. For a static spherically symmetric metric, use the area radius and proper-time normalization at infinity to solve $\text{Ric} = 0$, obtaining

$$g = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\Omega^2, \quad r_s = \frac{2GM}{c^2}.$$

Prove that its connected exterior isometry group is $\mathbb{R} \times SO(3)$.

Exercise 13.2 (Regge–Wheeler decomposition). Set $G = c = 1$ temporarily and define the tortoise coordinate by $dr_*/dr = (1 - 2M/r)^{-1}$. Separate the scalar wave equation using the $SO(3)$ decomposition and prove that $u_{\ell m} = r\phi_{\ell m}$ satisfies

$$\left(-\partial_t^2 + \partial_{r_*}^2 - V_\ell(r)\right) u_{\ell m} = 0, \quad V_\ell(r) = \left(1 - \frac{2M}{r}\right) \left(\frac{\ell(\ell+1)}{r^2} + \frac{2M}{r^3}\right).$$

Construct the conserved Wronskian and the ingoing/outgoing scattering coefficients, and prove their flux identity.

Exercise 13.3 (Confluent Heun functions and quasinormal modes). Define a confluent Heun function as the Frobenius solution at a regular singular point of

$$y'' + \left(\alpha + \frac{\beta+1}{z} + \frac{\gamma+1}{z-1}\right) y' + \left(\frac{\mu}{z} + \frac{\nu}{z-1}\right) y = 0,$$

with its parameters and initial exponent specified. Transform the separated Schwarzschild radial equation to this form and derive the three-term recurrence of its Frobenius coefficients. With time dependence $e^{-i\omega t}$, define a quasinormal frequency as a zero of the connection coefficient multiplying the incoming solution at infinity for the solution ingoing at the horizon. Express that coefficient as the limit of the recurrence's continued fraction and prove its convergence under the minimal-solution condition.

Exercise 13.4 (Schwarzschild scalar ringdown). For $\ell \rightarrow \infty$, locate the maximum of V_ℓ at the unstable photon orbit $r = 3M$. Define the parabolic-cylinder function $D_\nu(z)$ as the solution of $y'' + (\nu + \frac{1}{2} - z^2/4)y = 0$ normalized by $D_\nu(z) \sim z^\nu e^{-z^2/4}$ for $|\arg z| < 3\pi/4$. Derive its connection formula by analytically continuing

$$D_\nu(z) = \frac{e^{-z^2/4}}{\Gamma(-\nu)} \int_0^\infty e^{-zt-t^2/2} t^{-\nu-1} dt, \quad \Re \nu < 0,$$

and match these functions across the potential maximum to obtain, for fixed overtone n ,

$$\frac{GM\omega_{\ell n}}{c^3} = \frac{\ell + 1/2 - i(n + 1/2)}{3\sqrt{3}} + O(\ell^{-1}).$$

For the scalar $\ell = 2$ fundamental mode, truncate the confluent-Heun continued fraction with a rigorously bounded minimal-solution tail and obtain

$$\frac{GM\omega_{20}}{c^3} = 0.48364\dots - 0.09676\dots i.$$

Show that the predicted scalar waveform has oscillation frequency $\Re\omega/(2\pi)$ and damping time $1/|\Im\omega|$. Explain how the continued-fraction value supplies the finite- ℓ correction and compare it with the eikonal approximation. State that this is a test-field prediction, not the gravitational $\ell = 2$ mode, and identify the mass calibration and measured oscillation and damping times.

Part III

Hyperbolic Geometry

14 $SL_2(\mathbb{R})$

Exercise 14.1 (Hyperbolic isometries and the Casimir). Let $SL_2(\mathbb{R})$ act on $\mathbb{H} = \{x + iy : y > 0\}$ by Möbius maps. Prove that it preserves $ds^2 = (dx^2 + dy^2)/y^2$, identify the stabilizer of i with $SO(2)$, and derive the Iwasawa and Cartan decompositions. With the nonnegative convention define

$$\Delta_{\mathbb{H}} = -y^2(\partial_x^2 + \partial_y^2)$$

and show that it is the action of the quadratic Casimir on right- $SO(2)$ invariant functions.

Exercise 14.2 (Principal, complementary, and discrete series). Construct the spherical principal series by unitary induction from the upper triangular subgroup, the complementary series by its invariant kernel, and the holomorphic and antiholomorphic discrete series on weighted Bergman spaces. Determine their Casimir and $SO(2)$ spectra, including the signs of the discrete-series weights. Prove irreducibility by cyclicity of a K -finite vector and classify the irreducible unitary representations by the possible lowest/highest weight or continuous Casimir parameter.

Exercise 14.3 ($SL_2(\mathbb{R})$ Plancherel decomposition). Construct the standard intertwining operator between opposite principal series, compute its scalar on every $SO(2)$ type from the raising and lowering relations, and determine its poles and residues. Shift its contour to recover the discrete-series summands and derive the group Plancherel measure, with both even and odd principal-series densities. Prove inversion first for compactly supported smooth K -finite functions and extend it to L^2 .

Exercise 14.4 (Legendre functions and the Mehler–Fock transform). With fixed branches, use the Gauss hypergeometric function to define

$$P_{\nu}^{\mu}(z) = \frac{1}{\Gamma(1-\mu)} \left(\frac{z+1}{z-1} \right)^{\mu/2} {}_2F_1 \left(-\nu, \nu+1; 1-\mu; \frac{1-z}{2} \right).$$

Derive the associated Legendre equation and show that the spherical function at spectral parameter λ is $P_{-1/2+i\lambda}(\cosh r)$. From the radial Sturm–Liouville problem prove the Mehler–Fock inversion and Plancherel formulas, including the density $\lambda \tanh(\pi\lambda)$. Derive the analogous fixed- $SO(2)$ -type transform with $P_{-1/2+i\lambda}^{-m}$.

Exercise 14.5 (Hyperbolic-drum spectra). Let a membrane occupy a geodesic disk of curvature radius a and hyperbolic radius R , with wave speed v and Dirichlet boundary. Separate its wave equation into $SO(2)$ types and prove that its resonances are

$$\omega_{m,k} = \frac{v}{a} \sqrt{\frac{1}{4} + \lambda_{m,k}^2}, \quad P_{-1/2+i\lambda_{m,k}}^{-|m|}(\cosh R) = 0.$$

Prove the $m \neq 0$ double degeneracy and the Weyl asymptotic from the Mehler–Fock spectral measure. The zero locations, degeneracies, and area coefficient are the three measurable predictions.

15 $SL_2(\mathbb{Z})$

Exercise 15.1 (Modular group and its fundamental domain). Define $SL_2(\mathbb{Z})$, its action on $\mathbb{H}$, and $S\tau = -1/\tau$, $T\tau = \tau + 1$. Prove the presentation $PSL_2(\mathbb{Z}) = \langle S, T \mid S^2 = (ST)^3 = 1 \rangle$ by reducing points to the standard fundamental domain. Compute its hyperbolic area and the cusp, elliptic-point, and orbifold stabilizer data.

Exercise 15.2 (Eisenstein series and the scattering coefficient). Construct $L^2(SL_2(\mathbb{Z}) \backslash \mathbb{H})$ with hyperbolic measure and the nonnegative Laplacian. Define the Eisenstein series $E(\tau, s) = \sum_{\Gamma_\infty \backslash \Gamma} (\Im \gamma \tau)^s$, where Γ_∞ is the stabilizer of the cusp at infinity, for $\Re s > 1$. Define the zeta function $\zeta(s) = \sum_{n \geq 1} n^{-s}$ there and its completion $\xi(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$. Unfold the Eisenstein series, continue it using the theta Mellin transform, and prove that its constant term is

$$E(x + iy, s)_0 = y^s + \varphi(s) y^{1-s}, \quad \varphi(s) = \frac{\xi(2s-1)}{\xi(2s)}.$$

Derive its functional equation and $\varphi(s)\varphi(1-s) = 1$, including unitarity on $\Re s = 1/2$.

Exercise 15.3 (Automorphic spectral and trace decomposition). Develop the Friedrichs realization of the nonnegative Laplacian at the cusp. Truncate the Eisenstein series, prove the Maass–Selberg relation, and obtain the orthogonal decomposition into constants, cusp forms, and the continuous Eisenstein integral. Build the automorphic kernel from a compactly supported point-pair invariant, unfold it by conjugacy classes, and derive the Selberg trace identity with its identity, elliptic, parabolic, and hyperbolic terms. Recover the Weyl area term.

Exercise 15.4 (Modular forms, theta constants, and the eta function). Define modular forms by their weight, multiplier, and cusp growth. Starting from the theta transformation proved in Fourier Analysis, define $\vartheta_2, \vartheta_3, \vartheta_4$ by their shifted or signed lattice sums and derive all their S, T transformations without re-proving Poisson summation. Define $\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n)$ with $q = e^{2\pi i \tau}$, and derive its transformations from the Jacobi triple product. Construct E_4, E_6 , prove the valence formula, and show

$$\mathbb{C}[E_4, E_6] = \bigoplus_{k \geq 0} M_k, \quad \Delta = \eta^{24} = \frac{E_4^3 - E_6^2}{1728}.$$

Exercise 15.5 (Eisenstein scattering on the modular surface). Model a scalar waveguide by the modular surface with its cusp attached to a single asymptotic lead. At spectral value $s = \frac{1}{2} + ir$ with $r > 0$, so that the nonnegative Laplacian has value $\frac{1}{4} + r^2$, identify $y^{1/2-ir}$ and $y^{1/2+ir}$ as the two cusp waves. Orient the lead so that the y^s coefficient is the unit incoming amplitude and the y^{1-s} coefficient is outgoing, and prove that the reflection amplitude is $\varphi(\frac{1}{2} + ir)$. Write $\varphi(\frac{1}{2} + ir) = e^{2i\delta(r)}$ and, for wave speed v and $\omega(r) = v\sqrt{\frac{1}{4} + r^2}$, derive the Wigner delay

$$\tau_W(r) = 2 \frac{d\delta}{d\omega} = \frac{2\sqrt{\frac{1}{4} + r^2}}{vr} \frac{d\delta}{dr}.$$

Show that the meromorphically continued poles of φ are the scattering resonances. Thus the completed-zeta quotient predicts the reflected phase, time delay, and resonance locations; separate the constant-curvature, single-cusp, scalar-wave, and lead-matching assumptions from those measured quantities.

Part IV

Conformal Field Theory

16 Virasoro Algebra

Exercise 16.1 (The Virasoro central extension). Let $\text{Diff}^+(S^1)$ act on the circle and on scalar fields by pullback, and identify its infinitesimal action with smooth vector fields. After complexification on the punctured complex line let $L_n = -z^{n+1}\partial_z$, $n \in \mathbb{Z}$, be the distinguished Fourier vector fields. Prove $[L_m, L_n] = (m - n)L_{m+n}$. Define

$$\omega(L_m, L_n) = \frac{m^3 - m}{12} \delta_{m+n, 0}$$

and prove directly that it is alternating and satisfies the Lie-algebra cocycle identity. Define $\text{Vir} = \bigoplus_{n \in \mathbb{Z}} \mathbb{C}L_n \oplus \mathbb{C}C$ by

$$[L_m, L_n] = (m - n)L_{m+n} + \omega(L_m, L_n)C, \quad [C, L_n] = 0,$$

and verify the involution $L_n^* = L_{-n}$, $C^* = C$. Recover the cocycle as

$$\omega(f\partial_z, g\partial_z) = \frac{1}{12} \text{Res}_{z=0}(f'''(z)g(z)dz),$$

compute its change under a new circle coordinate as a coboundary, and deduce that the central-extension class is coordinate-independent.

Exercise 16.2 (Positive-energy highest-weight decomposition). Call a unitary Virasoro representation positive-energy when C acts by a real scalar c , L_0 has spectrum bounded below with finite-dimensional eigenspaces, and the algebraic sum of those eigenspaces is dense. Use $[L_0, L_n] = -nL_n$ and the involution to prove that every irreducible such representation is generated by a vector v satisfying

$$L_0v = hv, \quad L_nv = 0 \ (n > 0), \quad Cv = cv,$$

and prove the orthogonal decomposition of an arbitrary positive-energy representation into irreducible highest-weight summands by induction over its energy levels. Define the Verma module $M(c, h)$ by this universal highest-weight property. Apply the PBW theorem to show that its level- N subspace is indexed by partitions of N ; write $p(N)$ for their number and put $p(N) = 0$ for $N < 0$. Derive

$$\text{ch } M(c, h) = \text{Tr}_{M(c, h)} q^{L_0 - c/24} = \frac{q^{h - (c-1)/24}}{\eta(\tau)}, \quad q = e^{2\pi i \tau}.$$

Exercise 16.3 (Shapovalov determinants and unitary representations). Give $M(c, h)$ the contravariant form determined by $\langle v, v \rangle = 1$ and $L_n^* = L_{-n}$. Compute its Gram determinants at levels one and two. Put

$$c = 1 - 6(b - b^{-1})^2, \quad h_{r,s}(b) = \frac{(r/b - sb)^2 - (b^{-1} - b)^2}{4}.$$

Using the PBW filtration, construct a singular vector when $h = h_{r,s}(b)$, compare the degree and leading coefficient of the level- N determinant as a polynomial in h , and prove

$$\det_N(c, h) = C_N \prod_{\substack{r, s \geq 1 \\ rs \leq N}} (h - h_{r,s}(b))^{p(N - rs)}$$

with $C_N > 0$ fixed by the ordered PBW monomials. Track the signs between consecutive zero curves and quotient the null spaces to prove that irreducible positive-energy modules are unitary for $c \geq 1$, $h \geq 0$, and, below $c = 1$, precisely at

$$c_m = 1 - \frac{6}{m(m+1)}, \quad h_{r,s}^{(m)} = \frac{((m+1)r - ms)^2 - 1}{4m(m+1)}$$

with $m \geq 2$, $1 \leq r \leq m-1$, $1 \leq s \leq m$, modulo $(r, s) \sim (m-r, m+1-s)$.

Exercise 16.4 (Minimal-model characters). For $K \in \mathbb{N}$ and $\mu \in \mathbb{Z}$ define the level theta series

$$\Theta_{\mu,K}(\tau) = \sum_{n \in \mathbb{Z}} q^{(2Kn+\mu)^2/(4K)}.$$

Determine the inclusions among the singular Verma submodules at $(c_m, h_{r,s}^{(m)})$, resolve the irreducible quotient by their alternating chain, and take its Euler character to prove

$$\chi_{r,s}^{(m)}(\tau) = \frac{\Theta_{(m+1)r-ms, m(m+1)}(\tau) - \Theta_{(m+1)r+ms, m(m+1)}(\tau)}{\eta(\tau)}.$$

Use the Poisson summation formula to derive the finite S - and T -matrices of these characters. At $m = 3$ identify the three weights $0, 1/2, 1/16$ and prove

$$\chi_0 = \frac{1}{2} \left(\sqrt{\frac{\vartheta_3}{\eta}} + \sqrt{\frac{\vartheta_4}{\eta}} \right), \quad \chi_{1/2} = \frac{1}{2} \left(\sqrt{\frac{\vartheta_3}{\eta}} - \sqrt{\frac{\vartheta_4}{\eta}} \right), \quad \chi_{1/16} = \frac{1}{\sqrt{2}} \sqrt{\frac{\vartheta_2}{\eta}},$$

fixing the square roots by their positive q -expansions.

Exercise 16.5 (Virasoro characters and monstrous moonshine). Define a vertex operator algebra as a positive-energy Virasoro representation with vacuum $\mathbf{1}$, translation operator D , conformal vector, and state-field map $Y(a, z) = \sum_{n \in \mathbb{Z}} a_{(n)} z^{-n-1}$ satisfying

$$Y(\mathbf{1}, z) = 1, \quad Y(a, z)\mathbf{1} \in a + zV[[z]], \quad [D, Y(a, z)] = \partial_z Y(a, z),$$

and $(z-w)^N[Y(a, z), Y(b, w)] = 0$ for sufficiently large N , with the conformal field equal to $T(z)$. Derive the mode Jacobi identity and prove that the graded dimension is a Virasoro character. Define $j = E_4^3/\Delta$ and use the valence formula to show that every modular function holomorphic away from the cusp is a polynomial in j . Compute

$$J(\tau) = j(\tau) - 744 = q^{-1} + 196884q + 21493760q^2 + \dots$$

Write $J(q) = \sum_{n \geq -1} c(n)q^n$. Construct the graded traces of the moonshine vertex operator algebra and prove its Borcherds denominator identity

$$p^{-1} \prod_{m>0, n \in \mathbb{Z}} (1 - p^m q^n)^{c(mn)} = J(p) - J(q).$$

Extract the denominator recursion and use the vacuum normalization and first graded pieces to prove that its identity character is J . Verify $196884 = 1 + 196883$ and $21493760 = 1 + 196883 + 21296876$ as decompositions into irreducible Monster dimensions.

Exercise 16.6 (Conformal spectra and the Cardy prediction). Define a unitary two-dimensional conformal theory on a circle of circumference L by commuting positive-energy Virasoro actions with central charges $c, \bar{c}$ and weights $h, \bar{h}$, and use units $\hbar = 1$ in conformal finite-size formulas. Define $T(z) = \sum_n L_n z^{-n-2}$ and transform it from the plane to the cylinder, deriving the Schwarzian contribution from the Virasoro cocycle. Obtain the Hamiltonian and momentum

$$H = \frac{2\pi v}{L} \left(L_0 + \bar{L}_0 - \frac{c + \bar{c}}{24} \right), \quad P = \frac{2\pi}{L} (L_0 - \bar{L}_0).$$

For a finite character family and nonnegative integral multiplicity matrix M , prove that the torus trace $Z = \sum_{i,j} M_{ij} \chi_i \bar{\chi}_j$ is modular invariant exactly when M commutes with its S, T matrices; identify the integral-spin and $c - \bar{c} \in 24\mathbb{Z}$ conditions imposed by T . If $h_{\min}$ is the least chiral weight coupled to the vacuum, define $c_{\text{eff}} = c - 24h_{\min}$. Apply the S transformation and a saddle-point estimate to prove

$$\log \rho(\Delta) = 2\pi \sqrt{\frac{c_{\text{eff}} \Delta}{6}} + o(\sqrt{\Delta}).$$

Thus predict both the finite-size ground-energy shift $-\pi v(c + \bar{c})/(12L)$ and the high-energy level-counting rate, specifying v, L , and the measured energies.

17 Ising Model

Exercise 17.1 ($\mathbb{Z}_2$ spin symmetry and the partition function). For a finite graph $\Gamma = (V, E)$ and spins $\sigma : V \rightarrow \{\pm 1\}$ define

$$Z_\Gamma(K, h) = \sum_{\sigma} \exp \left(K \sum_{ij \in E} \sigma_i \sigma_j + h \sum_{i \in V} \sigma_i \right).$$

Identify the global $\mathbb{Z}_2$ action at $h = 0$. Expand each edge factor into even and odd parts and prove the high-temperature loop expansion. Expand about the two ordered configurations to obtain the dual domain-wall expansion. On the square lattice derive Kramers–Wannier duality $\sinh(2K) \sinh(2K^*) = 1$, including all finite-torus boundary sectors.

Exercise 17.2 (Transfer-matrix decomposition). For an $M \times N$ periodic square lattice, construct the row Hilbert space as the tensor product of the two-dimensional spin object and define the positive transfer operator $T_M(K)$. Prove $Z_{M,N} = \text{Tr}(T_M^N)$ and that T_M commutes with global spin reversal and translation. Introduce the Clifford algebra with the fixed convention $\{\gamma_a, \gamma_b\} = 2\delta_{ab}$, construct the Jordan–Wigner operators, and reduce T_M in each spin structure to commuting two-dimensional momentum blocks. Derive their rapidities

$$\cosh \gamma(k) = \cosh(2K^*) \cosh(2K) - \sinh(2K^*) \sinh(2K) \cos k.$$

Exercise 17.3 (Onsager free energy). Put $\beta = (k_B T)^{-1}$. Take the thermodynamic limit of the largest transfer eigenvalue and derive the isotropic free energy

$$-\beta f(K) = \frac{1}{2} \log(2 \sinh 2K) + \frac{1}{2\pi} \int_0^\pi \text{arcosh}(\cosh 2K \coth 2K - \cos \theta) d\theta.$$

Locate its only nonanalytic point at $K_c = \frac{1}{2} \log(1 + \sqrt{2})$ and derive the logarithmic heat-capacity singularity.

Exercise 17.4 (Toeplitz determinants and spontaneous magnetization). Express a long-distance spin correlation as a Toeplitz determinant. Derive the scalar symbol by the Clifford decomposition, factor it into inside and outside analytic parts, and prove the strong Szegő limit needed for its ordered phase. Obtain

$$M(K) = \lim_{r \rightarrow \infty} \langle \sigma_0 \sigma_r \rangle^{1/2} = \left(1 - \sinh^{-4}(2K)\right)^{1/8}, \quad K > K_c.$$

Exercise 17.5 (Virasoro characters of the Ising transfer matrix). Use the convention $q = e^{2\pi i \tau}$ from $SL_2(\mathbb{Z})$. Take the scaling limit of the four transfer-matrix spin structures and factor their fermionic mode products by the Jacobi triple product. Identify the resulting three functions with the $c = 1/2$ Virasoro characters $\chi_0, \chi_{1/2}, \chi_{1/16}$. Match the lattice vacuum, energy, and spin sectors to the weights $0, 1/2, 1/16$, derive their coefficient multiplicities, and use their S, T matrices to construct the modular-invariant torus partition function $|\chi_0|^2 + |\chi_{1/2}|^2 + |\chi_{1/16}|^2$.

Exercise 17.6 (Ising criticality). Restore $K = J/(k_B T)$. Predict

$$k_B T_c = \frac{2J}{\log(1 + \sqrt{2})}, \quad M(T) \sim C(T_c - T)^{1/8}.$$

On a cylinder of circumference L , use the three Ising characters and the conformal Hamiltonian to derive the universal ground-energy correction $-\pi v_s/(12L)$ and excitation gaps $(2\pi v_s/L)(h + \bar{h} + n + \bar{n})$. State separately the square-lattice, nearest-neighbor, zero-field assumptions; the measured temperature, magnetization, and gaps; and the nonuniversal calibrations J and v_s .

Part V

Quantum Algebra

18 Affine Kac–Moody Algebras

Exercise 18.1 (Central extensions of loop algebras). Let K_c be a simply connected compact Lie group whose complexified Lie algebra is the complex simple Lie algebra $\mathfrak{g}$. Define its smooth loop group $LK_c = C^\infty(S^1, K_c)$ and let it act on the loop-field configuration space LK_c by left multiplication, with S^1 acting by rotation of the source circle; prove that these actions combine to an $LK_c \rtimes S^1$ action. Normalize the invariant form $(\ , \)$ so that long roots have squared length two. Define the polynomial loop algebra $L\mathfrak{g} = \mathfrak{g} \otimes \mathbb{C}[z, z^{-1}]$ and

$$\widehat{\mathfrak{g}} = L\mathfrak{g} \oplus \mathbb{C}K \oplus \mathbb{C}d$$

by

$$[x \otimes z^m, y \otimes z^n] = [x, y] \otimes z^{m+n} + m\delta_{m+n,0}(x, y)K, \quad [d, x \otimes z^n] = nx \otimes z^n.$$

Obtain the central term from the residue cocycle $\text{Res}_{z=0}(\partial_z x(z), y(z)) dz$ and prove the Jacobi identity. For K_c construct the corresponding $U(1)$ -central extension of its smooth loop group at an integer level k by exponentiating the cocycle. If ϑ is the left Maurer–Cartan form, define the Cartan three-form as the scalar multiple of $(\vartheta, [\vartheta, \vartheta])$ that integrates to one on every long-root $SU(2)$ subgroup, and prove independence of the filling disk from its integral cohomology class.

Exercise 18.2 (Integrable highest-weight decomposition). Extend a Cartan algebra of $\mathfrak{g}$ by K and d . Define the null root by $\delta(d) = 1$ and by zero on the other Cartan summands. With highest root θ , set $\alpha_0 = \delta - \theta$ and $\alpha_0^\vee = K - \theta^\vee$, and define the affine Weyl group, level, and dominant affine weights. Apply the rank-one $\mathfrak{sl}_2$ strings for every affine simple root to prove that an irreducible highest-weight module is integrable at level $k \in \mathbb{N}$ exactly when its finite highest weight λ satisfies

$$\langle \lambda, \theta^\vee \rangle \leq k.$$

Call a unitary loop-group representation positive-energy when the rotation energy $-d$ has spectrum bounded below with finite-dimensional eigenspaces. Prove that every such representation decomposes orthogonally into these modules. For $\widehat{\mathfrak{sl}}_2$ identify the level- k summands $L_0, L_1, \dots, L_k$.

Exercise 18.3 (Sugawara Virasoro decomposition). Let $\mathcal{C}_{\mathfrak{g}} \in \mathfrak{g} \otimes \mathfrak{g}$ be the inverse tensor to the invariant form and define the dual Coxeter number $h^\vee$ by the scalar $2h^\vee$ through which the associated Casimir acts on $\mathfrak{g}$. Write $J_{x,n}$ for the action of $x \otimes z^n$ and set the linear map $\mathcal{N}_{p,q} : \mathfrak{g} \otimes \mathfrak{g} \rightarrow \text{End}(L)$ to send $x \otimes y$ to $J_{x,p}J_{y,q}$ for $p < 0$ and to $J_{y,q}J_{x,p}$ for $p \geq 0$. Define the Sugawara operators intrinsically by contraction with $\mathcal{C}_{\mathfrak{g}}$:

$$L_n = \frac{1}{2(k + h^\vee)} \sum_{m \in \mathbb{Z}} \mathcal{N}_{n-m,m}(\mathcal{C}_{\mathfrak{g}}).$$

Prove from the affine commutators that these operators satisfy the Virasoro relations and

$$c = \frac{k \dim \mathfrak{g}}{k + h^\vee}, \quad h_\lambda = \frac{(\lambda, \lambda + 2\rho)}{2(k + h^\vee)}.$$

For $\widehat{\mathfrak{sl}}_2$ derive $c = 3k/(k + 2)$ and $h_\ell = \ell(\ell + 2)/(4(k + 2))$.

Exercise 18.4 (Weyl–Kac characters). Put $y = e^{2\pi iz}$ and, without changing the modular nome convention, extend the level theta series by

$$\Theta_{m,K}(z, \tau) = \sum_{n \in \mathbb{Z}} q^{K(n+m/(2K))^2} y^{K(n+m/(2K))}.$$

Define $\chi_\ell^{(k)}(z, \tau) = \text{Tr}_{L_\ell}(q^{L_0 - c/24} y^{H/2})$, where H is the $\mathfrak{sl}_2$ coroot operator. Construct the affine BGG resolution by reflections in the walls of the fundamental alcove and take its Euler character to prove the Weyl–Kac formula. For $\widehat{\mathfrak{sl}}_2$ obtain

$$\chi_\ell^{(k)}(z, \tau) = \frac{\Theta_{\ell+1,k+2}(z, \tau) - \Theta_{-\ell-1,k+2}(z, \tau)}{\Theta_{1,2}(z, \tau) - \Theta_{-1,2}(z, \tau)}, \quad 0 \leq \ell \leq k.$$

Use Poisson summation to prove that their continuous values at $z = 0$ transform by

$$S_{\ell m} = \sqrt{\frac{2}{k+2}} \sin \frac{(\ell+1)(m+1)\pi}{k+2}, \quad T_{\ell \ell} = e^{2\pi i(h_\ell - c/24)}.$$

Derive orthogonality and completeness of this finite sine transform.

Exercise 18.5 (Knizhnik–Zamolodchikov conformal blocks). For finite-dimensional $\mathfrak{g}$ -modules $V_1, \dots, V_n$, let Ω_{ij} be the invariant Casimir contraction on the i th and j th factors. On the configuration space of distinct points define

$$\nabla_{KZ} = d - \frac{1}{k + h^\vee} \sum_{i < j} \Omega_{ij} d \log(z_i - z_j).$$

Prove the infinitesimal braid relations for the Ω_{ij} and hence flatness. For $k \geq 2$ and four fundamental $\mathfrak{sl}_2$ modules, decompose the invariant tensors through the two intermediate-spin channels, place three points at $0, 1, \infty$, and reduce the horizontal-section equation to the Gauss hypergeometric equation. Derive its Euler integral from the Gamma integral, use it to obtain the connection matrix between the 0 and 1 channels, and compute the two braid eigenvalues. Record the resulting braid representation by its eigenvalue ratio and channel-connection matrix.

Exercise 18.6 (Wess–Zumino–Witten finite-size spectroscopy). Define the diagonal $SU(2)_k$ Wess–Zumino–Witten state space as $\bigoplus_{\ell=0}^k L_\ell \otimes \overline{L}_\ell$. For a periodic critical system whose scaling limit has this state space, use the Sugawara Hamiltonian and the affine characters to derive

$$E_0(L) = e_\infty L - \frac{\pi v}{6L} \frac{3k}{k+2} + o(L^{-1}),$$

$$E_{\ell, N, \bar{N}}(L) - E_0(L) = \frac{2\pi v}{L} (2h_\ell + N + \bar{N}) + o(L^{-1}),$$

where e_∞ is the bulk ground-energy density and $N, \bar{N} \in \mathbb{Z}_{\geq 0}$ are descendant levels, with every degeneracy given by the corresponding character coefficients. For an antiferromagnetic spin-1/2 Heisenberg chain of N_s sites, set the lattice spacing to one, take $k = 1$, $L = N_s$, and $v = \pi J/2$ to predict the leading primary gap $\pi^2 J/(2N_s)$ and ground-energy correction $-\pi^2 J/(12N_s)$. Separate the WZW scaling hypothesis, periodic boundary condition, velocity calibration, and logarithmic finite-size corrections from the measured gaps and degeneracies.

19 Quantum Groups

Exercise 19.1 (The Hopf algebra $U_v(\mathfrak{sl}_2)$). For $v \in \mathbb{C}^\times$ not a root of unity, define $U_v(\mathfrak{sl}_2)$ by

$$KEK^{-1} = v^2 E, \quad KFK^{-1} = v^{-2} F, \quad [E, F] = \frac{K - K^{-1}}{v - v^{-1}},$$

with

$$\Delta K = K \otimes K, \quad \Delta E = E \otimes K + 1 \otimes E, \quad \Delta F = F \otimes 1 + K^{-1} \otimes F,$$

and construct its counit and antipode from the Hopf identities. Define $[n]_v = (v^n - v^{-n})/(v - v^{-1})$ and construct every finite-dimensional type-one irreducible V_n from a highest-weight vector. Prove the deformed Clebsch–Gordan rule $V_m \otimes V_n \simeq \bigoplus_{j=0}^{\min(m, n)} V_{m+n-2j}$. For any ordered collection of type-one modules, use the iterated coproduct to define the diagonal $U_v(\mathfrak{sl}_2)$ action on their tensor-product state space and identify quantum-group-invariant observables with its commuting endomorphisms.

Exercise 19.2 (Universal R -matrix). Write $K = v^H$ formally and, in the completion acting on type-one modules, define

$$\mathcal{R} = v^{H \otimes H/2} \sum_{r \geq 0} v^{r(r-1)/2} \frac{(v - v^{-1})^r}{[r]_v!} E^r \otimes F^r.$$

Verify directly on highest-weight modules that $\mathcal{R}\Delta(a) = \Delta^{\text{op}}(a)\mathcal{R}$ and

$$(\Delta \otimes \text{id})\mathcal{R} = \mathcal{R}_{13}\mathcal{R}_{23}, \quad (\text{id} \otimes \Delta)\mathcal{R} = \mathcal{R}_{13}\mathcal{R}_{12}.$$

Define the braid group B_n by generators σ_i with the far-commutativity and braid relations; deduce its actions on tensor powers and the quantum Yang–Baxter equation. On the two fundamental channels, where the sum terminates and has a regular root-of-unity value, specialize at $v = e^{\pi i/(k+2)}$. Prove that the ratio of its two braid eigenvalues and its recoupling matrix agree with the framing-independent Knizhnik–Zamolodchikov braid data.

Exercise 19.3 (Root-of-unity tilting modules and ribbon quotients). Construct the divided-power integral form of $U_v(\mathfrak{sl}_2)$ over $\mathbb{Z}[v, v^{-1}]$ and specialize it at $v = e^{\pi i/(k+2)}$. Define Weyl modules, dual Weyl modules, and tilting modules through filtrations by both. Use the antipode, evaluation, and coevaluation to define their rigid duals and prove closure of the tilting subcategory under tensor products and duals. Construct the pivotal element from the square of the antipode and use it to define the categorical trace. Define a morphism to be negligible when its categorical trace vanishes after every composable morphism, prove that negligible morphisms form a tensor ideal, and form the Karoubi-completed quotient by formally adjoining the images of idempotents. Show that it is semisimple with simple objects $V_0, \dots, V_k$, quantum dimensions

$$d_a = \frac{\sin((a+1)\pi/(k+2))}{\sin(\pi/(k+2))},$$

and truncated Clebsch–Gordan products. Construct the ribbon element from the specialized R -matrix, define its compatibility with the coproduct, antipode, and pivotal element, prove the ribbon identities in the quotient, and derive

$$\theta_a = \exp\left(\frac{\pi i a(a+2)}{2(k+2)}\right).$$

Exercise 19.4 (Quantum Peter–Weyl decomposition). Define the restricted Hopf dual $\mathcal{O}_v(SL_2)$ as the span of matrix coefficients. Prove its algebraic Peter–Weyl decomposition

$$\mathcal{O}_v(SL_2) = \bigoplus_{n \geq 0} V_n^\vee \otimes V_n$$

and derive Schur orthogonality for the invariant positive functional when $0 < v < 1$, after equipping it with the compact real-form involution and writing the resulting Hopf $*$ -algebra as $\mathcal{O}_v(SU(2))$.

Exercise 19.5 (Basic hypergeometric and dual q -Hahn functions). Put $Q = v^2$ and keep this base distinct from the modular nome. For $n \in \mathbb{Z}_{\geq 0}$ and $|Q| < 1$ define

$$(a; Q)_n = \prod_{j=0}^{n-1} (1 - aQ^j), \quad (a; Q)_\infty = \prod_{j \geq 0} (1 - aQ^j), \quad (a_1, \dots, a_m; Q)_n = \prod_i (a_i; Q)_n,$$

and define

$${}_r\phi_s \left(\begin{matrix} a_1, \dots, a_r \\ b_1, \dots, b_s \end{matrix}; Q, z \right) = \sum_{k \geq 0} \frac{(a_1, \dots, a_r; Q)_k}{(Q, b_1, \dots, b_s; Q)_k} \left((-1)^k Q^{k(k-1)/2} \right)^{1+s-r} z^k.$$

Derive the Q -binomial and Q -Gauss sums by finite telescoping and analytic continuation, and show that terminating identities extend rationally in Q away from denominator zeros. Define the dual Q -Hahn polynomials by

$$R_n(Q^{-x} + \gamma\delta Q^{x+1}; \gamma, \delta, N \mid Q) = {}_3\phi_2 \left(\begin{matrix} Q^{-n}, Q^{-x}, \gamma\delta Q^{x+1} \\ \gamma Q, Q^{-N} \end{matrix}; Q, Q \right).$$

Derive their difference equation and finite orthogonality. Solve the $U_v(\mathfrak{sl}_2)$ lowering recurrence and identify its normalized Clebsch–Gordan coefficients with these polynomials after matching the three integer labels and normalization. Define the terminating Q -Racah family by

$$\mathcal{R}_n(Q^{-x} + \delta Q^{x-N}) = {}_4\phi_3 \left(\begin{matrix} Q^{-n}, \alpha\beta Q^{n+1}, Q^{-x}, \delta Q^{x-N} \\ \alpha Q, \beta\delta Q, Q^{-N} \end{matrix}; Q, Q \right)$$

and derive its finite orthogonality and recoupling identity from two associations of a fourfold tensor product.

Exercise 19.6 (Six-vertex transfer matrices and the XXZ chain). Write $v = e^\eta$ and define, with additive spectral parameter u ,

$$R(u) = \begin{pmatrix} \sinh(u + \eta) & 0 & 0 & 0 \\ 0 & \sinh u & \sinh \eta & 0 \\ 0 & \sinh \eta & \sinh u & 0 \\ 0 & 0 & 0 & \sinh(u + \eta) \end{pmatrix}.$$

Evaluate the finite universal R -matrix on $V_1 \otimes V_1$, prove that its checked form satisfies the Hecke quadratic relation, and derive the displayed spectral matrix by Baxterization. Prove its Yang–Baxter equation directly from the braid and Hecke relations. Form the periodic row transfer matrix, prove commutativity for different u , and take its logarithmic derivative at $u = 0$ to obtain, up to scalar and scale,

$$H_{XXZ} = \sum_{j=1}^N (\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y + \cosh \eta \sigma_j^z \sigma_{j+1}^z).$$

Develop the algebraic Bethe ansatz and derive the Bethe equations and energy in every fixed-magnetization sector.

Exercise 19.7 (XXZ multiplet spectroscopy). For an open $U_v(\mathfrak{sl}_2)$ -symmetric spin chain, decompose three adjacent spin sectors in the two recoupling orders. Express the change of channel through normalized dual Q -Hahn coefficients and its iterates through Q -Racah polynomials. For a weak probe transforming as V_2 , derive the allowed total-spin channels and their relative transition intensities as squares of these coefficients. For the periodic chain, use the Bethe roots to give the corresponding resonance frequencies. State boundary symmetry, anisotropy $\Delta = \cosh \eta$, polarization, temperature, and line-resolution assumptions separately from the predicted missing lines, intensity ratios, and frequencies.

20 Dunkl Operators

Exercise 20.1 (Reflection groups and commuting Dunkl operators). Let $\mathfrak{a}$ be a finite-dimensional Euclidean space, let $R \subset \mathfrak{a}$ be a reduced root system, and let W be the reflection group generated by $s_\alpha x = x - 2\langle \alpha, x \rangle \alpha / \langle \alpha, \alpha \rangle$. Choose R_+ and a nonnegative W -invariant multiplicity function $k : R \rightarrow \mathbb{R}_{\geq 0}$. For $\xi \in \mathfrak{a}$ define on polynomial functions

$$T_\xi f(x) = \partial_\xi f(x) + \sum_{\alpha \in R_+} k_\alpha \langle \alpha, \xi \rangle \frac{f(x) - f(s_\alpha x)}{\langle \alpha, x \rangle}.$$

Prove that the divided differences preserve polynomials, establish $wT_\xi w^{-1} = T_{w\xi}$, and prove $[T_\xi, T_\eta] = 0$ by reducing the commutator to the rank-two root subsystems. For

$$w_k(x) = \prod_{\alpha \in R_+} |\langle \alpha, x \rangle|^{2k_\alpha},$$

derive the integration-by-parts identity for the T_ξ on the complement of the reflection arrangement and extend it to Schwartz functions. Contract two copies of $\xi \mapsto T_\xi$ with the inverse Euclidean metric to define the Dunkl Laplacian. Restrict it to W -invariant functions and conjugate by $w_k^{1/2}$ to obtain the rational Calogero Hamiltonian, including its inverse-square coupling constants.

Exercise 20.2 (Dunkl kernels). Construct the degree-preserving intertwining operator V_k recursively from $T_\xi V_k = V_k \partial_\xi$ and $V_k 1 = 1$. Prove its uniqueness, its W -equivariance, and its convergence on exponential power series. Define the Dunkl kernel $E_k(x, y) = V_k(e^{\langle \cdot, y \rangle})(x)$ and prove

$$T_\xi^x E_k(x, y) = \langle \xi, y \rangle E_k(x, y), \quad E_k(x, y) = E_k(y, x).$$

Use the differential-difference equation and its reflection maximum principle to prove $|E_k(ix, y)| \leq 1$ for real x, y and $k \geq 0$, and derive the exponential growth bound needed for contour and integral arguments.

Exercise 20.3 (Dunkl transforms and generalized Bessel decomposition). With $c_k = \int_{\mathfrak{a}} e^{-|x|^2/2} w_k(x) dx$, define

$$\mathcal{F}_k f(y) = c_k^{-1} \int_{\mathfrak{a}} f(x) E_k(x, -iy) w_k(x) dx.$$

Construct the Dunkl heat kernel explicitly from E_k , prove its semigroup and approximate-identity properties, and use them to prove inversion and Plancherel. Show that $\mathcal{F}_k(T_\xi f) = i\langle \xi, y \rangle \mathcal{F}_k f$. Define the generalized Bessel kernel by

$$J_k(x, y) = \frac{1}{|W|} \sum_{w \in W} E_k(ix, wy),$$

and derive the corresponding transform on W -invariant functions. In rank one identify J_k with the normalized Bessel function of order $k - 1/2$, and derive its Hankel inversion formula directly from Dunkl inversion.

Exercise 20.4 (Heckman–Opdam Jacobi and Jack polynomials). Put $\rho_k = \frac{1}{2} \sum_{\alpha \in R_+} k_\alpha \alpha$. On the regular complex torus of the weight lattice define the trigonometric Dunkl operators

$$\mathcal{D}_\xi = \partial_\xi - \langle \rho_k, \xi \rangle + \sum_{\alpha \in R_+} k_\alpha \langle \alpha, \xi \rangle \frac{1 - s_\alpha}{1 - e^{-\alpha}}.$$

Verify their commutativity by the rank-two root-system calculation and derive the degenerate affine-Hecke cross relations that they satisfy. For each dominant weight λ , define the Heckman–Opdam Jacobi polynomial as the unique W -invariant orbit-sum-triangular joint eigenfunction of the W -invariant polynomials in the $\mathcal{D}_\xi$. Prove orthogonality and completeness for the compact-torus weight

$$\delta_k(x) = \prod_{\alpha \in R_+} |e^{\alpha(x)/2} - e^{-\alpha(x)/2}|^{2k_\alpha},$$

construct the multiplicity-shift intertwiners from the positive-root directional operators, compute their adjoints, derive the norm and product kernels. In rank one identify these functions with Jacobi polynomials, including the parameter correspondence.

For $R = A_{r-1}$ and constant multiplicity g , identify the compact Jacobi polynomials with Jack polynomials $P_\lambda^{(1/g)}(x_1, \dots, x_r)$. Derive their differential operator, orthogonality for $\prod_{i < j} |x_i - x_j|^{2g}$ on the unit torus, Cauchy kernel, norm, and Pieri rule, with every normalization obtained as a specialization of the preceding root-system construction.

Exercise 20.5 (Calogero–Sutherland spectroscopy). On the configuration space of r indistinguishable particles on a circle of radius R , let S_r act by permuting the angular positions and define the dimensionless operator

$$H_g = -\frac{1}{2} \sum_{i=1}^r \partial_{\theta_i}^2 + \frac{g(g-1)}{4} \sum_{i < j} \sin^{-2} \frac{\theta_i - \theta_j}{2}, \quad g > 0.$$

Select its self-adjoint realization in the permutation-invariant sector by the collision behavior $\Psi = O(|\theta_i - \theta_j|^g)$, and put $\mathcal{H}_g = \hbar^2 H_g / (mR^2)$. Conjugate H_g by $\Psi_0 = \prod_{i < j} |e^{i\theta_i} - e^{i\theta_j}|^g$ and identify the result with the type- A trigonometric Dunkl operator. Prove that its normalized eigenfunctions are $\Psi_0 P_\lambda^{(1/g)}$ and that

$$E_\lambda = \frac{1}{2} \sum_{i=1}^r \left(\lambda_i + g \left(\frac{r+1}{2} - i \right) \right)^2.$$

Use the Jack Pieri rule for the density mode $p_1 = \sum_i e^{i\theta_i}$ to derive the allowed transitions $\lambda \rightarrow \lambda + \square$ and their squared normalized intensities. Thus a periodic density probe predicts the angular frequencies $\omega_{\lambda\mu} = \hbar |E_\mu - E_\lambda| / (mR^2)$, missing lines, and intensity ratios; state the inverse-square interaction, identical-particle sector, radius, coupling, temperature, and spectral-resolution assumptions separately from those observables.

21 Double Affine Hecke Algebras

Exercise 21.1 (Double-affine braid groups). For the type- A_{r-1} root system in $\{x \in \mathbb{R}^r : \sum x_i = 0\}$, construct its Weyl chambers, affine reflection hyperplanes, and alcoves. Identify the affine braid group with braids in an annulus. Show that allowing every strand to wind around both cycles of a torus gives the double-affine braid group, with finite braid generators T_i and commuting winding families X_j and Y_j . Derive its braid and winding relations directly from homotopies of torus braids. Impose $(T_i - t^{1/2})(T_i + t^{-1/2}) = 0$ and fix the total winding commutator to q to define $\mathbf{H}_{q,t}(GL_r)$. Here q is a multiplicative difference parameter, not the modular nome; when this finite Hecke action is compared with Quantum Groups use $t^{1/2} = v$, and otherwise keep t and v independent. Filter near $q = 1$, compute the associated graded difference-reflection relations, and recover the trigonometric Dunkl algebra acting on the torus.

Exercise 21.2 (Polynomial representation and Cherednik operators). Let $\mathcal{T}_r = (\mathbb{C}^\times)^r$ be the algebraic torus and let

$$\mathcal{P}_r = \mathbb{C}(q, t^{1/2})[x_1^{\pm 1}, \dots, x_r^{\pm 1}]$$

be its Laurent-polynomial functions. Let X_i be multiplication, let s_i exchange x_i, x_{i+1} , define $T_{q,x_i} f(x_1, \dots, x_r) = f(x_1, \dots, qx_i, \dots, x_r)$, and put

$$T_i = t^{1/2} s_i + (t^{1/2} - t^{-1/2}) \frac{s_i - 1}{x_i/x_{i+1} - 1}, \quad (\omega f)(x) = f(qx_r, x_1, \dots, x_{r-1}),$$

$$Y_i = T_i \cdots T_{r-1} \omega T_1^{-1} \cdots T_{i-1}^{-1}.$$

Verify that these difference-reflection operators define the double-affine Hecke action on the function space of $\mathcal{T}_r$, including preservation of Laurent polynomials, the Hecke and affine braid relations, and $[Y_i, Y_j] = 0$. Prove their triangular action on monomials in Bruhat-refined dominance order. Show on symmetric Laurent polynomials that

$$t^{(r-1)/2} \sum_i Y_i = D_{q,t}, \quad D_{q,t} = \sum_i \prod_{j \neq i} \frac{tx_i - x_j}{x_i - x_j} T_{q,x_i}.$$

Exercise 21.3 (Macdonald polynomials and their kernel). For a partition λ of length at most r , define P_λ as the monic symmetric polynomial triangular in monomial symmetric functions and satisfying

$$D_{q,t} P_\lambda = \left(\sum_{i=1}^r q^{\lambda_i} t^{r-i} \right) P_\lambda.$$

For $0 < q, t < 1$, prove symmetry of $D_{q,t}$ for the torus weight

$$\Delta_{q,t}(x) = \prod_{i \neq j} \frac{(x_i/x_j; q)_\infty}{(tx_i/x_j; q)_\infty}$$

and deduce orthogonality and completeness. In the stable symmetric-function ring define the power-sum pairing

$$\langle p_\lambda, p_\mu \rangle_{q,t} = \delta_{\lambda\mu} z_\lambda \prod_j \frac{1 - q^{\lambda_j}}{1 - t^{\lambda_j}}.$$

Determine the Macdonald norm from the adjoint Pieri operators, define Q_λ to be dual to P_λ for this pairing, and prove the Cauchy kernel

$$\sum_\lambda P_\lambda(a; q, t) Q_\lambda(b; q, t) = \prod_{i,j} \frac{(ta_i b_j; q)_\infty}{(a_i b_j; q)_\infty}.$$

Construct nonsymmetric joint Y -eigenfunctions by intertwiners and recover P_λ with the finite-Hecke symmetrizer.

Exercise 21.4 (Elliptic deformation and Macdonald specialization). For $|p|, |q| < 1$ define

$$\theta(z; p) = (z; p)_\infty (p/z; p)_\infty, \quad \Gamma(z; p, q) = \prod_{j,k \geq 0} \frac{1 - p^{j+1} q^{k+1} / z}{1 - p^j q^k z}.$$

Prove their difference and reflection identities. Use the multiplicative theta addition formula to construct the rank-one elliptic difference-reflection operators and verify their dynamical braid relation. Derive the terminating Frenkel–Turaev sum by telescoping that relation. Take $p \rightarrow 0$ to obtain the Askey–Wilson operator and its ${}_4\phi_3$ eigenpolynomials; specialize its four boundary parameters to recover the rank-one Macdonald operator. Record every parameter map so that p is always the elliptic nome, q the difference step, and t the Hecke parameter.

Exercise 21.5 (Ruijsenaars–Schneider spectrum). Put $x_j = e^{i\theta_j}$ and regard the commuting symmetric polynomials in Y_i as quantum integrals of the compact trigonometric Ruijsenaars–Schneider system. With the positive Macdonald weight, construct their self-adjoint realization and prove

$$E_\lambda = \sum_{i=1}^r q^{\lambda_i} t^{r-i}$$

for the first normalized Hamiltonian. Derive the first Macdonald Pieri rule for multiplication by $x_1 + \cdots + x_r$ and hence the allowed transitions $\lambda \rightarrow \lambda + \square$ and their squared normalized intensities. Thus a periodic density probe predicts exact line positions, missing lines, and intensities; state the finite-particle, relativistic-difference coupling, boundary, and spectral-resolution assumptions.

Part VI

Algebraic Geometry

22 Algebraic Curves

Exercise 22.1 (Affine and projective curves). For a commutative unital ring A , define prime ideals, localization, the Zariski space $\text{Spec } A$, and its structure sheaf by the rings A_f on distinguished opens. Prove that every stalk is local and define a locally ringed space, morphisms of locally ringed spaces, affine schemes, and schemes by affine covers. Define Noetherian rings, Krull dimension, and a regular Noetherian local ring by $\dim_{A/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2) = \dim A$. Construct Proj from homogeneous prime ideals and homogeneous localization. Define a projective curve over $\mathbb{C}$ to be a smooth connected one-dimensional closed subscheme of a projective space, with smoothness expressed by regular local rings. Construct the analytification of a smooth projective curve from its local rings and prove it is a compact Riemann surface. Define divisors by valuations of the function field and prove that principal divisors have degree zero by the argument principle.

Exercise 22.2 (Homology, periods, and de Rham's theorem). Define singular chains, boundary, homology, cochains, cup product, orientations, and the fundamental class of an oriented closed manifold. For the differential forms developed with Hamiltonian actions, define integration over smooth singular simplices. Prove Stokes's theorem and show that integration gives a cochain map. Construct a good cover, compare the singular and de Rham Čech double complexes with partitions of unity, and prove

$$H_{\text{dR}}^*(M) \simeq H^*(M; \mathbb{R})$$

as graded rings. Derive Mayer–Vietoris and Poincaré duality from the integration pairing. For a compact Riemann surface, prove that the intersection form on H_1 is unimodular and alternating and construct a symplectic marking. Define its genus g by $\text{rank } H_1 = 2g$ and identify this integer with its topological handle number.

Exercise 22.3 (Elliptic curves and translation symmetry). For a nonsingular cubic $E : y^2 = 4x^3 - g_2x - g_3$ with discriminant $g_2^3 - 27g_3^2 \neq 0$, construct the group law by divisors and prove the chord–tangent formula. Show that the holomorphic form dx/y has no zeros and that its period map identifies

$$E(\mathbb{C}) \simeq \mathbb{C}/(\mathbb{Z}\omega_1 + \mathbb{Z}\omega_2).$$

Construct the Weierstrass function from its normally convergent lattice sum, prove $\wp'^2 = 4\wp^3 - g_2\wp - g_3$, and show that $z \mapsto (\wp(z), \wp'(z))$ is the inverse uniformization. Derive the addition formula from translation on E .

Exercise 22.4 (Branched covers and Riemann–Hurwitz). Define a branched cover of compact curves and its ramification divisor. Define the canonical divisor as the divisor of any nonzero meromorphic differential and prove that its divisor class is independent of that differential. Prove

Riemann–Hurwitz by comparing canonical divisors. For the normalization of $y^2 = \prod_{j=1}^{2g+1} (x - e_j)$, include the chart at infinity and prove smoothness and genus g .

Exercise 22.5 (Abel–Jacobi decomposition). Construct a symplectic marking of first homology, normalize the holomorphic differentials, and define the Abel map and Jacobian. Prove Abel’s theorem and Jacobi inversion by the argument principle and the Riemann bilinear relations.

Exercise 22.6 (Elliptic integrals and Jacobi functions). For $0 < k < 1$ define

$$K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - k^2 \sin^2 \theta}}, \quad E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \theta} d\theta.$$

Invert the Abel integral $u = \int_0^x dt / \sqrt{(1 - t^2)(1 - k^2 t^2)}$ on its elliptic curve to define $\operatorname{sn}(u; k)$ globally; define cn and dn by the algebraic relations with branches fixed at zero. Derive their differential equations, addition laws, periods, Fourier series, and the hypergeometric representation $K(k) = \frac{\pi}{2} {}_2F_1(1/2, 1/2; 1; k^2)$.

Exercise 22.7 (Nonlinear pendulum periods). For a pendulum of length L in uniform gravity g , use energy conservation and the elliptic Abel map to prove that oscillations of amplitude $0 < \theta_0 < \pi$ have period

$$T(\theta_0) = 4\sqrt{\frac{L}{g}} K\left(\sin \frac{\theta_0}{2}\right).$$

Expand the hypergeometric series to obtain

$$T = 2\pi\sqrt{\frac{L}{g}} \left(1 + \frac{\theta_0^2}{16} + \frac{11\theta_0^4}{3072} + O(\theta_0^6)\right)$$

and derive the logarithmic divergence at the separatrix. Identify L, g, θ_0 , and the measured return time, and distinguish ideal rigid, frictionless motion from experimental corrections.

23 Riemann–Roch

Exercise 23.1 (Line-bundle automorphisms and sheaf cohomology). Define a line bundle $p : L \rightarrow X$ and immediately define a section over $U \subset X$ as a map $s : U \rightarrow L$ with $p \circ s = \operatorname{id}_U$; write $\Gamma(U, L)$ for all such sections. Define the sheaf of sections, locally free and coherent sheaves, tensor product, dual, and pullback. Define $\operatorname{Aut}(X, L)$ to consist of pairs formed by an automorphism ϕ of X and a fiberwise-linear bundle isomorphism $\tilde{\phi} : L \rightarrow L$ covering it, and prove that $(\phi, \tilde{\phi})s = \tilde{\phi} \circ s \circ \phi^{-1}$ acts on sections and functorially on every sheaf-cohomology group. Construct Čech cohomology from an open cover. Prove the affine vanishing theorem for quasi-coherent sheaves directly from localization, deduce that affine covers compute their sheaf cohomology, prove independence of the chosen affine cover by the refinement double complex, and derive the long exact sequence from a short exact sequence of sheaves. Identify divisors modulo principal divisors with the Picard group.

Exercise 23.2 (Serre duality and Riemann–Roch). For a smooth projective curve X , construct the residue pairing

$$H^1(X, L) \otimes H^0(X, K_X \otimes L^{-1}) \longrightarrow \mathbb{C}$$

from Čech representatives and prove it is perfect. Prove Riemann–Roch by induction on a divisor using the exact sequence at a point:

$$h^0(L) - h^1(L) = \deg L + 1 - g.$$

Define the holomorphic Euler characteristic $\chi(X, L) = \sum_q (-1)^q \dim H^q(X, L)$. Deduce $\deg K_X = 2g - 2$, the vanishing theorem for $\deg L > 2g - 2$, and the dimension of meromorphic functions with prescribed poles.

Exercise 23.3 (Dolbeault cohomology and Serre duality). Define a complex manifold by a holomorphic atlas and a holomorphic vector bundle by its locally free sheaf of sections. Decompose complex differential forms into bidegrees and define ∂ and $\bar{\partial}$. Prove the $\bar{\partial}$ -Poincaré lemma, construct the fine Dolbeault resolution, and identify coherent sheaf cohomology of a holomorphic bundle E with $H_{\bar{\partial}}^{0,q}(X, E)$. On a compact complex manifold with Hermitian metric, construct the formal adjoint and Dolbeault Laplacian from the metric volume form, establish elliptic regularity from local Fourier estimates and a partition of unity, and prove the harmonic-representative theorem. Use the Hodge star and integration pairing to prove the perfect duality

$$H^q(X, E) \otimes H^{d-q}(X, K_X \otimes E^\vee) \longrightarrow \mathbb{C}$$

for $d = \dim_{\mathbb{C}} X$, recovering the curve theorem as the case $d = 1$.

Exercise 23.4 (Bott–Borel–Weil for SL_2). Identify $SL_2(\mathbb{C})/B \simeq \mathbb{P}^1$ and compute

$$H^q(\mathbb{P}^1, \mathcal{O}(n)) = \begin{cases} \text{Sym}^n(\mathbb{C}^2)^\vee, & q = 0, n \geq 0, \\ \text{Sym}^{-n-2}(\mathbb{C}^2), & q = 1, n \leq -2, \\ 0, & \text{otherwise,} \end{cases}$$

using Riemann–Roch and Serre duality. Recover the shifted reflection $s \cdot n = -n - 2$.

Exercise 23.5 (Bott–Borel–Weil for reductive groups). For general reductive G , cover G/B by Bruhat cells, filter its Čech complex by Bruhat order, identify the associated graded terms with Verma modules, and prove that singular $\lambda + \rho$ gives zero while regular $\lambda + \rho$ gives

$$H^{\ell(w)}(G/B, L_\lambda) \simeq V_{w(\lambda+\rho)-\rho}^\vee$$

for the unique w carrying it to the dominant chamber.

Exercise 23.6 (Twisted differential operators on $\mathbb{P}^1$). Derive from the Möbius action on $\mathbb{P}^1$ the operators on $\Gamma(\mathbb{A}^1, \mathcal{O}(n))$

$$e = \partial_z, \quad h = -2z\partial_z + n, \quad f = -z^2\partial_z + nz,$$

and verify the $\mathfrak{sl}_2$ relations and invariance of polynomials of degree at most n . Define $\Omega = \frac{1}{2}h^2 + ef + fe$, compute its scalar, and define the resulting infinitesimal character. Construct the twisted differential-operator sheaf $\mathcal{D}_n$ by gluing the displayed chart operators.

Exercise 23.7 (Beilinson–Bernstein localization). For a weight λ , define $\mathcal{D}_\lambda$ by twisting differential operators with L_λ . Construct the order filtration, identify its associated graded with functions on $T^*(G/B)$, and use the moment map and the Koszul resolution of its fibers to prove

$$\Gamma(G/B, \mathcal{D}_\lambda) \simeq U(\mathfrak{g})/U(\mathfrak{g}) \ker \chi_\lambda.$$

For regular dominant λ , prove exactness and faithfulness of global sections by the Bruhat filtration and deduce the localization equivalence. Recover Borel–Weil from the localized highest-weight module.

Exercise 23.8 (Bergman kernels). Give X a Hermitian area form and L a positive Hermitian holomorphic line bundle. Define the orthogonal projection $L^2(X, L) \rightarrow H^0(X, L)$ by its intrinsic integral kernel $B_L(x, y)$. Prove its reproducing, Hermitian, and projection identities. Define Λ as contraction with the area form. Derive the extremal characterization $B_L(x, x) = \sup\{|s(x)|^2 : \|s\|_{L^2} = 1\}$ and, from rescaling the heat kernel of the Dolbeault Laplacian, prove for L^k

$$B_{L^k}(x, x) = \frac{k}{2\pi} i\Lambda F_L + O(1)$$

uniformly. On $\mathbb{P}^1$ and on a complex torus compute the kernel explicitly, reusing rather than redefining the earlier theta functions.

Exercise 23.9 (Landau-level degeneracy and density). Define a magnetic line bundle for a charge q on a compact Riemann surface to be a positive Hermitian holomorphic line bundle L . Define its Chern connection as the unique unitary connection whose $(0, 1)$ part is $\bar{\partial}_L$, prove existence and uniqueness, and require its curvature to be $F_L = -iqB/\hbar$, so that $\frac{q}{2\pi\hbar} \int_X B = N \in \mathbb{Z}$. Define its unitary gauge group as the unitary bundle automorphisms over id_X , prove that it acts on the quantum state space, and show that automorphisms in $\text{Aut}(X, L)$ preserving the metric and connection commute with the magnetic Hamiltonian. Factor its magnetic Laplacian through the Dolbeault operator and identify the lowest Landau level with $H^0(X, L)$. When $N > 2g - 2$, prove

$$\dim H^0(X, L) = N + 1 - g, \quad \rho(x) = B_L(x, x).$$

For $X = \mathbb{P}^1$ and $L = \mathcal{O}(N)$ with its invariant metric, use the $SU(2)$ action, Bott–Borel–Weil, and the Bergman projection identity to identify the state representation with $\text{Sym}^N(\mathbb{C}^2)^\vee$ and prove the constant density $\rho = (N + 1)/\text{area}(X)$. Thus predict both the exact number of lowest-level states and their local particle density; state flux quantization, curvature, boundary, interaction, and field-uniformity assumptions.

24 Finite-Gap Spectral Curves

Exercise 24.1 (KdV Lax flows). Let the translation circle act on the phase space $\mathcal{U} = C^\infty(\mathbb{R}/a\mathbb{Z}, \mathbb{R})$ of periodic potentials by pullback. On periodic smooth functions define $L = -\partial_x^2 + u$ and $P = -4\partial_x^3 + 3(u\partial_x + \partial_x u)$. Compute the commutator and prove that $L_t = [P, L]$ is

$$u_t - 6uu_x + u_{xxx} = 0.$$

Prove that the Lax vector field commutes with translations. Construct the monodromy of $L\psi = E\psi$, prove that its discriminant is constant under the flow, and show that the periodic spectrum is preserved. Derive the commuting odd operator for a stationary higher KdV flow and the Burchall–Chaundy polynomial relation.

Exercise 24.2 (Hyperelliptic spectral curves). Normalize the projective closure of $y^2 = \prod_{j=0}^{2g} (E - E_j)$ and include its smooth chart at infinity. Choose real ordered branch points and construct the real ovals. Normalize the holomorphic differentials by a symplectic homology marking and prove from the bilinear relations that the period matrix is symmetric with positive imaginary part. Define

$$\Theta(z \mid \Omega) = \sum_{n \in \mathbb{Z}^g} e^{\pi i n^T \Omega n + 2\pi i n^T z}$$

and derive its quasi-periodicity. State and prove the symplectic change of marking formula and the invariance of the reconstructed potential below.

Exercise 24.3 (Baker–Akhiezer functions). At the point over $E = \infty$, put $k = \sqrt{E}$. Construct normalized second-kind differentials with principal parts $d(ik)$ and $d(4ik^3)$ and period vectors U, V . By Riemann–Roch prove uniqueness of the function with a prescribed nonspecial pole divisor and essential factor $e^{ikx+4ik^3t}$. Derive its theta formula and show directly that it solves $L\psi = E\psi$ and $\psi_t = P\psi$. Compare the coefficient at infinity to prove

$$u(x, t) = c - 2\partial_x^2 \log \Theta(Ux + Vt + Z \mid \Omega)$$

and hence the stated sign convention for KdV.

Exercise 24.4 (Bloch–Floquet decomposition). For a real periodic potential of period a , define the translation by a and decompose $L^2(\mathbb{R})$ over its $U(1)$ characters. Prove that $\Delta(E) = \text{Tr } M(E)$ satisfies $\Delta(E) = 2\cos(ka)$ on a Bloch wave and that the spectrum is exactly $\{E : |\Delta(E)| \leq 2\}$. Show that simple branch points of the spectral curve are band edges and that closing all but g gaps is equivalent to the finite-genus polynomial relation above.

Exercise 24.5 (Lamé band-gap spectroscopy). For

$$L_k = -\frac{d^2}{dx^2} + 2k^2 \text{sn}^2(x; k)$$

of period $2K(k)$, use the Jacobi identities to prove that dn , cn , and sn have band edge energies $k^2, 1, 1 + k^2$. Prove

$$\text{spec}(L_k) = [k^2, 1] \cup [1 + k^2, \infty).$$

After restoring a spatial scale κ and factor $\hbar^2/(2m)$, predict the two band widths and the gap width $\hbar^2\kappa^2 k^2/(2m)$. State how Bloch spectroscopy measures the three edge energies and distinguish the one-gap periodic-potential model from lattice, interaction, and finite-size corrections.

Part VII

Algebraic Topology

25 Index Theory

Exercise 25.1 (Homotopy, CW complexes, and vector bundles). Define pointed homotopy, homotopy equivalence, loop and suspension spaces, homotopy groups, and CW complexes. Prove cellular approximation and the long exact homotopy sequence of a fibration. Define a topological complex vector bundle intrinsically by local product structure and fiberwise linear operations. Construct the infinite Grassmannian from its finite-dimensional subspace functor and prove that homotopy classes $[X, BU(n)]$ classify rank- n bundles over a finite CW complex. Define $K^0(X)$ by group completion under direct sum, prove homotopy invariance and the splitting principle, and derive its reduced suspension and clutching descriptions. When X is smooth, define a smooth vector bundle and prove by approximation and partitions of unity that every topological bundle is isomorphic to one with compatible smooth structure.

Exercise 25.2 ($U(1)$ gauge symmetry and characteristic classes). For the smooth vector bundles just constructed, define connections, curvature, and gauge transformations intrinsically. Prove that the gauge group acts on the affine space of connections and by conjugation on every naturally

coupled elliptic operator, preserving its spectrum and index. From invariant polynomials and the Bianchi identity prove that $\text{Tr exp}(iF/2\pi)$ is closed and connection-independent in cohomology. Construct integral Chern classes from transition functions and prove that their real images agree with Chern–Weil forms. Use the splitting principle to derive the Chern character and Todd and $\hat{A}$ classes and prove their multiplicative identities.

Exercise 25.3 (Fredholm operators and elliptic symbols). Define Fredholm operators and their analytic index, prove stability under compact perturbations, and construct the symbol class of an elliptic operator in $K_c^0(T^*M)$. Construct the bounded transform and prove that its index depends only on the compactly supported symbol class.

Exercise 25.4 (Bott periodicity and the topological index). Construct the harmonic-oscillator Clifford symbol on a complex vector space, compute its one-dimensional kernel, and prove that external product with its class gives the Bott isomorphism by radial contraction and Mayer–Vietoris. Construct Thom isomorphisms by the same Clifford class and define the topological index by embedding M in Euclidean space. Prove independence of the embedding by tubular neighborhoods and isotopy.

Exercise 25.5 (Atiyah–Singer index equality). Prove equality of analytic and topological indices first for the Bott operator, then for external products and open excision, and finally for every compact manifold by embedding and retraction. Apply the splitting principle and Chern–Weil theory to deduce

$$\text{ind } \bar{\partial}_E = \int_M \text{ch}(E) \text{Td}(T^{1,0}M)$$

on a compact complex manifold, and derive the $\hat{A}$ formula for a spin Dirac operator.

Exercise 25.6 (K3 elliptic genus). Define a K3 surface as a simply connected compact complex surface with trivial canonical bundle. Apply the Dolbeault index formula to the trivial bundle, use Serre duality, and prove $\int_{K3} c_2 = 24$ and $\hat{A}(K3) = 2$. With $y = e^{2\pi iz}$ and $q = e^{2\pi i\tau}$, define its elliptic genus by

$$y^{-1} \chi \left(K3, \bigotimes_{n \geq 1} \Lambda_{-yq^{n-1}} T^* \otimes \Lambda_{-y^{-1}q^n} T \otimes S_{q^n} T^* \otimes S_{q^n} T \right).$$

Here Λ_t and S_t are the total exterior- and symmetric-power operations in completed K -theory. Extend the theta constants to Jacobi theta functions by inserting $e^{2\pi i(n+a)z}$ in their characteristic lattice sums, and thereby define $\vartheta_1(z; \tau)$. Put

$$\phi_{0,1}(z, \tau) = 4 \left[\frac{\vartheta_2(z; \tau)^2}{\vartheta_2(0; \tau)^2} + \frac{\vartheta_3(z; \tau)^2}{\vartheta_3(0; \tau)^2} + \frac{\vartheta_4(z; \tau)^2}{\vartheta_4(0; \tau)^2} \right].$$

Use Riemann–Roch to derive the Chern-root theta product. Expand it, integrate, and prove $\text{Ell}_{K3}(z, \tau) = 2\phi_{0,1}(z, \tau)$. Derive its first Fourier–Jacobi coefficients from the theta identities already established.

Exercise 25.7 (Monopole harmonics). Let $L_N \rightarrow S^2$ be the homogeneous line bundle associated with the weight N character of $U(1) \subset SU(2)$. Use Peter–Weyl, rather than polar gauges, to decompose $L^2(S^2, L_N)$ into the $SU(2)$ representations V_j , $j = |N|/2, |N|/2 + 1, \dots$. Define the monopole harmonics as the corresponding distinguished matrix coefficients and derive their orthogonality, completeness, addition kernel, and magnetic-Laplacian values

$$\lambda_j = \frac{1}{R^2} \left(j(j+1) - \frac{N^2}{4} \right).$$

Identify the lowest level and its degeneracy $|N| + 1$ with Riemann–Roch.

Exercise 25.8 (Integer quantum Hall conductance). For a gapped family of Fermi projections $P(k)$ over the Brillouin torus, derive the Kubo response from adiabatic perturbation theory and prove

$$\sigma_{xy} = \frac{e^2}{h} \frac{i}{2\pi} \int_{\mathbb{T}^2} \text{Tr}(P \, dP \wedge dP) = C \frac{e^2}{h}, \quad C \in \mathbb{Z}.$$

Relate C to spectral flow by applying the index theorem to a flux insertion. For filled monopole-harmonic Landau levels, compute C from the associated line bundle and predict the Hall plateau and level degeneracy. Separate the gap, noninteracting-band, zero-temperature, and disorder-localization assumptions from the measured transverse conductance.

26 Cobordism

Exercise 26.1 (Spectra and stable homotopy). Using pointed CW complexes and the loop-suspension adjunction already developed, construct spectra and stable homotopy groups by iterated suspension, prove the Freudenthal stable range, and identify ordinary homology as maps into Eilenberg–Mac Lane spectra. Construct generalized homology and cohomology from a spectrum and prove their exactness and suspension axioms.

Exercise 26.2 (SO , Spin, and U structures). Construct the orthogonal frame bundle of a Riemannian manifold with its free right $O(d)$ action. Define a tangential G -structure as a lift of the stable tangent classifier through $BG \rightarrow BO$, equivalently as the appropriate reduction of this frame symmetry. Construct orientation, spin, and stable complex structures as the cases SO , Spin, and U , and identify their obstructions w_1 , w_2 , and the stable complex lifting condition. Define G -bordism by compact manifolds with compatible boundary structure and prove it is a graded ring under disjoint union and product.

Exercise 26.3 (Higher categories and bordisms). Define an (∞, n) -category as a complete n -fold Segal space: state the Segal composition maps, completeness condition, and the requirement that k -morphisms above level n be invertible. Define symmetric monoidal structures by commutative algebra objects in this model and construct the (∞, n) -category whose objects are compact 0-manifolds and whose k -morphisms are collared k -bordisms, diffeomorphisms, isotopies, and higher isotopies. Prove associativity of gluing from contractibility of collar choices and identify disjoint union as its symmetric tensor product.

Exercise 26.4 (Pontryagin–Thom decomposition). For an embedding of a closed G -manifold in Euclidean space, collapse the complement of a tubular neighborhood and obtain a map to the Thom space of its normal bundle. Prove independence under stabilization and bordism. Conversely, make a stable map transverse to the zero section and recover a G -manifold. Prove the isomorphism between G -bordism and the stable homotopy groups of the Thom spectrum MG . Derive complex orientations and the formal group law of MU from tensor product of line bundles.

Exercise 26.5 (Fully extended field theories). Define adjoints for k -morphisms in a symmetric monoidal (∞, n) -category and call an object fully dualizable when it has evaluation and coevaluation at every level with all required adjoints. Construct framed bordisms by handles and prove that evaluation on a point gives an equivalence between fully extended framed topological field theories and fully dualizable objects. Add homotopy fixed-point data for a tangential G -structure. Verify in dimension one that the snake identities defining finite-dimensional vector spaces at the beginning of the manuscript are exactly the required point data.

Exercise 26.6 (Eta functions and the APS boundary correction). For a self-adjoint elliptic operator D with discrete nonzero eigenvalues, define

$$\eta_D(s) = \sum_{\lambda \neq 0} \text{sign}(\lambda) |\lambda|^{-s}$$

for $\Re s$ large. Use the heat kernel Mellin transform to continue it to $s = 0$ and prove regularity there. Attach a cylinder to a compact manifold with boundary, analyze separated boundary modes, and prove

$$\text{ind } D_{\text{APS}}^+ = \int_M \widehat{A}(TM) \text{ch}(E) - \frac{\eta_{D_\partial}(0) + \dim \ker D_\partial}{2}.$$

For the circle operator with holonomy $e^{2\pi i a}$ derive $\eta(0) = 1 - 2a$ for $0 < a < 1$ from the Hurwitz-zeta functional equation.

Exercise 26.7 (Invertible phases and eta-response predictions). Define an invertible reflection-positive field theory as a symmetric monoidal map from the appropriate bordism category to invertible complex lines, and use stacking to form its classification group. Apply Pontryagin–Thom to identify deformation classes with maps from the relevant Thom spectrum to the Picard spectrum of invertible phases, including the symmetry structure in the tangential data. For a massive fermion whose boundary operator is D_∂ , derive the relative partition phase

$$Z_{\text{top}} = \exp \left[-\frac{\pi i}{2} (\eta_{D_\partial}(0) + \dim \ker D_\partial) \right].$$

Use the circle calculation to predict its change under one unit of flux and the resulting protected boundary spectral flow. State locality, gap, symmetry, and interaction assumptions separately from the measurable phase and crossing parity.

Part VIII

Tensor Categories

27 Geometric Satake

Exercise 27.1 (Loop groups and Schubert cells). For a complex reductive group G , define $G((z))$, $G[[z]]$, the faithfully flat quasi-compact topology on affine schemes, and sheafification of a functor of points. Define the fpqc quotient $\text{Gr}_G = G((z))/G[[z]]$ and prove that $G((z))$ acts transitively with stabilizer $G[[z]]$. For the induced $G[[z]]$ action, prove the Cartan decomposition

$$\text{Gr}_G = \bigsqcup_{\lambda \in X_*(T)_+} G[[z]] z^\lambda$$

and compute the orbit dimension $\langle 2\rho, \lambda \rangle$. Construct the Schubert varieties by orbit closure and derive their dominance order and convolution correspondence from modification of G -bundles on a formal disk.

Exercise 27.2 (Constructible derived categories and Verdier duality). For a finite stratification of a locally compact space, define bounded constructible complexes of sheaves and chain homotopies. Localize the homotopy category at quasi-isomorphisms to construct the constructible derived

category. Resolve by injective and soft sheaves to construct Rf_* , $Rf_!$, Lf^* , $f^!$, $\otimes^L$, and $R\mathcal{H}om$; derive their adjunctions, the projection formula, and proper base change from the corresponding statements for the resolutions. Construct the dualizing complex from compactly supported cochains and define Verdier duality $\mathbb{D}\mathcal{F} = R\mathcal{H}om(\mathcal{F}, \omega_X)$. Prove biduality and the interchange of Rf_* with $Rf_!$ under $\mathbb{D}$.

Exercise 27.3 (Perverse sheaves and convolution). For the Schubert stratification, define the perverse truncation by the support and cosupport inequalities. Prove that intersection cohomology extensions are simple perverse sheaves. Define convolution through the proper multiplication correspondence, prove perversity by semismallness, and derive associativity, commutativity from fusion on two moving points, and rigid duals from inversion and Verdier duality.

Exercise 27.4 (Satake fiber functor). Let Sat_G be the category of $G[[z]]$ -equivariant perverse sheaves on Gr_G . Prove exactness and faithfulness of total cohomology by the weight-functor decomposition over semi-infinite orbits. Use convolution to make it a symmetric monoidal fiber functor. Construct the group of its tensor automorphisms, recover its coordinate Hopf algebra from matrix coefficients and rigid duals, and prove the Tannaka equivalence with representations of a reductive group.

Exercise 27.5 (Geometric Satake equivalence). Compute the roots of this Tannaka group from the rank-one Schubert varieties and prove that its root datum is dual to that of G . Deduce the monoidal equivalence

$$\text{Sat}_G \simeq \text{Rep}(G^\vee)$$

and identify intersection cohomology of $\overline{\text{Gr}}_\lambda$ with the irreducible highest-weight representation V_λ .

Exercise 27.6 (Hall–Littlewood polynomials). For a partition λ define

$$P_\lambda(x; t) = \frac{1}{v_\lambda(t)} \sum_{w \in S_r} w \left(x^\lambda \prod_{i < j} \frac{x_i - tx_j}{x_i - x_j} \right)$$

with $v_\lambda(t)$ fixed by monicity. Prove polynomiality, triangularity, orthogonality, the Cauchy identity, and the Pieri rule. Over a finite field, stratify the convolution fibers in Gr_{GL_r} and show that their point counts give the Hall–Littlewood structure constants at $t = |\mathbb{F}|^{-1}$. At $t = 0$ recover the Weyl characters.

Exercise 27.7 (Gauge fields and 't Hooft defects). Choose a compact form K of the complex reductive group G above and a compatible maximal torus $T_K \subset K$ whose complexification is T . Define a principal K -bundle $P \rightarrow M$ over an oriented Riemannian four-manifold as a smooth fiber bundle with a free transitive right K -action on each fiber. Define a gauge field as a K -equivariant horizontal distribution on P , equivalently a connection form A satisfying $R_k^* A = \text{Ad}(k^{-1}) A$ and $A(\xi_P) = \xi$, where ξ_P is the fundamental vertical vector field generated by $\xi \in \mathfrak{k}$. Define its curvature by $F_A = dA + \frac{1}{2}[A \wedge A]$ and a gauge transformation as a K -equivariant bundle automorphism over id_M . Derive the induced action on A and F_A and prove gauge invariance of the Yang–Mills action

$$S(A) = \frac{1}{2e^2} \int_M \langle F_A \wedge *F_A \rangle$$

for an $\text{Ad } K$ -invariant inner product on $\mathfrak{k}$.

For an oriented embedded curve $C \subset M$, define an 't Hooft defect of charge $\lambda \in X_*(T)$ by requiring a Dirac-type singularity on every small two-sphere linking C : its reductions to T_K over the northern and southern hemispheres are glued on the equator by $e^{i\theta} \mapsto \lambda(e^{i\theta})$. Treat this reduction and its

singular asymptotics, rather than merely the isomorphism class of the underlying K -bundle, as the boundary condition. Prove that conjugate cocharacters give gauge-equivalent defects and that every charge has a unique dominant representative modulo the Weyl group.

Define the formal holomorphic local model of this boundary condition on a transverse disk with parameter z to be a G -bundle on $\text{Spec } \mathbb{C}[[z]]$ together with a trivialization over $\text{Spec } \mathbb{C}((z))$. Show that its transition function lies in $G((z))$, that changes of regular trivialization act by $G[[z]]$, and that the modification of charge λ lies in the Schubert orbit of z^λ in Gr_G . Show that two ordered defects on the disk determine two successive modifications, so bringing their support together gives precisely the convolution correspondence constructed above. State separately the compact-gauge-group, topological–holomorphic reduction, and point-defect idealization assumptions in this local model.

Exercise 27.8 ('t Hooft-line fusion). Model the protected magnetic line operators constructed above by Sat_G , and let L_λ be the Grothendieck class of the intersection-cohomology sheaf of $\overline{\text{Gr}}_\lambda$. Use the convolution description of collision and geometric Satake to prove in the protected fusion ring that

$$L_\lambda L_\mu = \sum_\nu \dim \text{Hom}_{G^\vee}(V_\nu, V_\lambda \otimes V_\mu) L_\nu.$$

Interpret the coefficient as the dimension of the corresponding space of local junction operators. Compute it from Hall–Littlewood multiplication at $t = 0$ and specialize to $G = PGL_2$, with V_n the irreducible $SU(2)$ -representation of highest weight n , to obtain

$$L_m L_n = \sum_{j=0}^{\min(m,n)} L_{m+n-2j}.$$

Thus predict the allowed magnetic charges, the unit multiplicity of every allowed junction channel, and the forbidden parity and triangle-inequality channels. State how defect-correlation measurements resolve these channels, and separate the protected multiplicities from coupling-dependent operator normalizations and from corrections to the local-defect idealization.

28 Fusion Categories

Exercise 28.1 (Rigid and unitary fusion categories). Define a unitary fusion category as a finite semisimple rigid $\mathbb{C}$ -linear C^* -tensor category with simple tensor unit. Define evaluation, coevaluation, pivotal and spherical structures, quantum dimensions, and the global dimension. Derive the snake, rigidity, Frobenius reciprocity, and $\sum_c d_{ab}^c d_c = d_a d_b$ identities intrinsically. Define braidings, twists, and nondegeneracy through the categorical trace of double braiding. Show that convolution categories of topological line defects furnish examples, while noninvertible simple objects require no underlying group. For simple objects $a_1, \dots, a_n, b$, define the fusion state space

$$\mathcal{H}(a_1, \dots, a_n; b) = \text{Hom}(b, a_1 \otimes \dots \otimes a_n).$$

Use $f^*g = \langle f, g \rangle 1_b$ to make it a Hilbert space. Prove that associators and braidings induce unitary maps among these fusion spaces and derive the resulting braid-group action and its invariance under isotopy.

Exercise 28.2 (Temperley–Lieb categories and fusion semisimplification). Define TL_d by planar pairings and loop value d , and prove the tensor and $*$ structures by gluing diagrams. Construct the Jones–Wenzl idempotents recursively and derive $f_1 \otimes f_n \simeq f_{n-1} \oplus f_{n+1}$. At $d = 2 \cos(\pi/(k+2))$,

identify the negligible ideal generated by f_{k+1} and prove that the semisimplification has simples $0, \dots, k$ and fusion

$$a \otimes b = \bigoplus_{\substack{c=|a-b| \pmod{2} \\ c \leq \min(a+b, 2k-a-b)}} c.$$

Construct the ribbon functor to the root-of-unity tilting quotient developed in Quantum Groups by sending one strand to V_1 , prove that its kernel is the negligible ideal generated by f_{k+1} , and identify the two semisimplifications.

Exercise 28.3 (Factorization homology and the Levin–Wen model). Define a disk algebra in a symmetric monoidal higher category and construct factorization homology by the colimit over embedded disks. Prove excision by cofinality of collar gluings. Apply it to the three-dimensional Turaev–Viro theory of a fusion category $\mathcal{C}$ and identify the category assigned to a circle with its Drinfeld center. Triangulate a closed surface, define the Levin–Wen vertex and plaquette projectors from the F -symbols, and prove their commutativity by the pentagon identity. Show that the ground-state space is the corresponding factorization-homology state space. Verify explicitly that $\mathcal{C} = SU(2)_1$ is the semion category and its Levin–Wen theory is doubled semion, not the toric code.

Exercise 28.4 (Modular S -matrix and the Verlinde decomposition). For $\mathcal{C} = SU(2)_k$, compute the categorical trace of double braiding from the root-of-unity ribbon structure constructed in Quantum Groups and recover the finite sine transform obtained from affine characters:

$$S_{ab} = \sqrt{\frac{2}{k+2}} \sin\left(\frac{(a+1)(b+1)\pi}{k+2}\right), \quad \theta_a = e^{\pi i a(a+2)/(2(k+2))}.$$

Prove directly that the two constructions agree, then use its orthogonality to deduce nondegeneracy of braiding and the Verlinde formula

$$N_{ab}^c = \sum_x \frac{S_{ax} S_{bx} \overline{S_{cx}}}{S_{0x}}.$$

Recover the truncated fusion rule and compute the torus ground-state degeneracy. Relate these trigonometric special functions to the root-of-unity limit of the quantum-group characters without redefining them.

Exercise 28.5 (Anyon interferometry). For simple anyons a, b , prove that a probe a encircling total charge b has normalized monodromy

$$M_{ab} = \frac{S_{ab} S_{00}}{S_{0a} S_{0b}}.$$

In a balanced two-path interferometer with controllable phase ϕ , derive

$$P_{\pm}(b) = \frac{1}{2} \left(1 \pm \Re(e^{i\phi} M_{ab}) \right).$$

When $\mathcal{C}$ is modular, identify the simple charges of the Drinfeld center realized by its Levin–Wen model with pairs and prove

$$S_{(a,b),(c,d)}^{Z(\mathcal{C})} = S_{ac} \overline{S_{bd}}.$$

Insert the $SU(2)_k$ sine matrix into the chiral and doubled monodromy formulas to predict every fringe displacement and visibility and show which charges are distinguishable by a fixed probe. State coherence, braiding adiabaticity, charge-superselection, path balance, and detector-resolution assumptions separately from these probabilities.

References

- [1] Y. Ai, L. Kong, and H. Zheng, *Topological orders and factorization homology*, Advances in Theoretical and Mathematical Physics 21 (2017), 1845–1894. doi:10.4310/ATMP.2017.v21.n8.a1.
- [2] M. Ando, M. J. Hopkins, and N. P. Strickland, *Elliptic spectra, the Witten genus and the theorem of the cube*, Inventiones Mathematicae 146 (2001), 595–687. doi:10.1007/s002220100175.
- [3] M. J. Hopkins, *Topological modular forms, the Witten genus, and the theorem of the cube*, Proceedings of the International Congress of Mathematicians, Zürich 1994, vol. 1, Birkhäuser, 1995, 554–565.
- [4] M. F. Atiyah, *K-Theory*, W. A. Benjamin, 1967.
- [5] M. F. Atiyah and I. M. Singer, *The index of elliptic operators I*, Annals of Mathematics 87 (1968), 484–530. doi:10.2307/1970715.
- [6] M. F. Atiyah and I. M. Singer, *The index of elliptic operators IV*, Annals of Mathematics 93 (1971), 119–138. doi:10.2307/1970756.
- [7] M. F. Atiyah, V. K. Patodi, and I. M. Singer, *Spectral asymmetry and Riemannian geometry I*, Mathematical Proceedings of the Cambridge Philosophical Society 77 (1975), 43–69. doi:10.1017/S0305004100049410.
- [8] M. F. Atiyah, V. K. Patodi, and I. M. Singer, *Spectral asymmetry and Riemannian geometry III*, Mathematical Proceedings of the Cambridge Philosophical Society 79 (1976), 71–99. doi:10.1017/S0305004100052105.
- [9] D. Ayala and J. Francis, *Factorization homology of topological manifolds*, Journal of Topology 8 (2015), 1045–1084. doi:10.1112/jtopol/jtv028.
- [10] A. A. Beilinson, J. Bernstein, and P. Deligne, *Faisceaux pervers*, Astérisque 100, Société Mathématique de France, 1982.
- [11] A. A. Beilinson and J. Bernstein, *Localisation de $\mathfrak{g}$ -modules*, Comptes rendus de l’Académie des sciences, Série I 292 (1981), 15–18.
- [12] E. D. Belokolos, A. I. Bobenko, V. Z. Enolskii, A. R. Its, and V. B. Matveev, *Algebro-Geometric Approach to Nonlinear Integrable Equations*, Springer, 1994.
- [13] R. E. Borcherds, *Monstrous moonshine and monstrous Lie superalgebras*, Inventiones Mathematicae 109 (1992), 405–444. doi:10.1007/BF01232032.
- [14] R. Bott and L. W. Tu, *Differential Forms in Algebraic Topology*, Graduate Texts in Mathematics 82, Springer, 1982.
- [15] R. Bott, *Homogeneous vector bundles*, Annals of Mathematics 66 (1957), 203–248. doi:10.2307/1969996.
- [16] S. Chandrasekhar, *The Mathematical Theory of Black Holes*, Clarendon Press, 1983.
- [17] X. Chen, Z.-C. Gu, Z.-X. Liu, and X.-G. Wen, *Symmetry protected topological orders and the group cohomology of their symmetry group*, Physical Review B 87 (2013), 155114. doi:10.1103/PhysRevB.87.155114.

- [18] I. Cherednik, *Double Affine Hecke Algebras*, London Mathematical Society Lecture Note Series 319, Cambridge University Press, 2005.
- [19] J. B. Conway, *A Course in Functional Analysis*, 2nd ed., Graduate Texts in Mathematics 96, Springer, 1990.
- [20] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Grundlehren der mathematischen Wissenschaften 290, Springer, 1999.
- [21] I. Corwin, *The Kardar–Parisi–Zhang equation and universality class*, Random Matrices: Theory and Applications 1 (2012), 1130001. doi:10.1142/S2010326311300014.
- [22] P. Deift and E. Trubowitz, *Inverse scattering on the line*, Communications on Pure and Applied Mathematics 32 (1979), 121–251. doi:10.1002/cpa.3160320202.
- [23] P. Di Francesco, P. Mathieu, and D. Sénéchal, *Conformal Field Theory*, Graduate Texts in Contemporary Physics, Springer, 1997.
- [24] C. L. Douglas, C. Schommer-Pries, and N. Snyder, *Dualizable tensor categories*, Memoirs of the American Mathematical Society 268 (2020), no. 1308. doi:10.1090/memo/1308.
- [25] T. Eguchi, H. Ooguri, A. Taormina, and S.-K. Yang, *Superconformal algebras and string compactification on manifolds with $SU(n)$ holonomy*, Nuclear Physics B 315 (1989), 193–221. doi:10.1016/0550-3213(89)90454-9.
- [26] P. Etingof, D. Nikshych, and V. Ostrik, *On fusion categories*, Annals of Mathematics 162 (2005), 581–642. doi:10.4007/annals.2005.162.581.
- [27] J. D. Fay, *Theta Functions on Riemann Surfaces*, Lecture Notes in Mathematics 352, Springer, 1973.
- [28] G. B. Folland, *Harmonic Analysis in Phase Space*, Princeton University Press, 1989.
- [29] G. B. Folland, *Real Analysis: Modern Techniques and Their Applications*, 2nd ed., Wiley, 1999.
- [30] O. Forster, *Lectures on Riemann Surfaces*, Graduate Texts in Mathematics 81, Springer, 1981.
- [31] D. S. Freed and M. J. Hopkins, *Reflection positivity and invertible topological phases*, Geometry & Topology 25 (2021), 1165–1330. doi:10.2140/gt.2021.25.1165.
- [32] W. Fulton and J. Harris, *Representation Theory: A First Course*, Graduate Texts in Mathematics 129, Springer, 1991.
- [33] G. H. Hardy and S. Ramanujan, *Asymptotic formulae in combinatory analysis*, Proceedings of the London Mathematical Society (2) 17 (1918), 75–115. doi:10.1112/plms/s2-17.1.75.
- [34] D. Johnson, *Spin structures and quadratic forms on surfaces*, Journal of the London Mathematical Society (2) 22 (1980), 365–373. doi:10.1112/jlms/s2-22.2.365.
- [35] A. Kapustin and N. Saulina, *The algebra of Wilson–’t Hooft operators*, Nuclear Physics B 814 (2009), 327–365. doi:10.1016/j.nuclphysb.2009.01.025.
- [36] M. Kashiwara, *The Riemann–Hilbert problem for holonomic systems*, Publications of the Research Institute for Mathematical Sciences 20 (1984), 319–365. doi:10.2977/PRIMS/1195181610.

- [37] R. Hartshorne, *Algebraic Geometry*, Graduate Texts in Mathematics 52, Springer, 1977.
- [38] F. Hirzebruch, *Topological Methods in Algebraic Geometry*, 3rd ed., Grundlehren der mathematischen Wissenschaften 131, Springer, 1966. doi:10.1007/978-3-642-62018-8.
- [39] L. Hörmander, *The Analysis of Linear Partial Differential Operators III*, Grundlehren der mathematischen Wissenschaften 274, Springer, 1985.
- [40] H. Iwaniec, *Spectral Methods of Automorphic Forms*, 2nd ed., Graduate Studies in Mathematics 53, American Mathematical Society, 2002.
- [41] V. G. Kac, *Infinite-Dimensional Lie Algebras*, third edition, Cambridge University Press, 1990.
- [42] A. Y. Kitaev, *Unpaired Majorana fermions in quantum wires*, Physics-Uspekhi 44 (2001), 131–136. doi:10.1070/1063-7869/44/10S/S29.
- [43] A. Kitaev, *Periodic table for topological insulators and superconductors*, AIP Conference Proceedings 1134 (2009), 22–30. doi:10.1063/1.3149495.
- [44] A. W. Knap, *Representation Theory of Semisimple Groups: An Overview Based on Examples*, Princeton University Press, 1986.
- [45] T. H. Koornwinder, *Askey–Wilson polynomials for root systems of type BC*, Contemporary Mathematics 138 (1992), 189–204. doi:10.1090/conm/138/1199128.
- [46] V. E. Korepin, N. M. Bogoliubov, and A. G. Izergin, *Quantum Inverse Scattering Method and Correlation Functions*, Cambridge University Press, 1993.
- [47] G. Kuperberg, *Another proof of the alternating-sign matrix conjecture*, International Mathematics Research Notices 1996, no. 3, 139–150. doi:10.1155/S1073792896000128.
- [48] H. B. Lawson, Jr. and M.-L. Michelsohn, *Spin Geometry*, Princeton University Press, 1989.
- [49] M. A. Levin and X.-G. Wen, *String-net condensation: A physical mechanism for topological phases*, Physical Review B 71 (2005), 045110. doi:10.1103/PhysRevB.71.045110.
- [50] J. Lurie, *On the classification of topological field theories*, Current Developments in Mathematics 2008 (2009), 129–280. doi:10.4310/CDM.2008.v2008.n1.a3.
- [51] G. W. Mackey, *Induced representations of locally compact groups I*, Annals of Mathematics 55 (1952), 101–139. doi:10.2307/1969423.
- [52] J. E. Marsden and T. S. Ratiu, *Introduction to Mechanics and Symmetry*, 2nd ed., Texts in Applied Mathematics 17, Springer, 1999.
- [53] I. Mirković and K. Vilonen, *Geometric Langlands duality and representations of algebraic groups over commutative rings*, Annals of Mathematics 166 (2007), 95–143. doi:10.4007/annals.2007.166.95.
- [54] R. G. Newton, *Scattering Theory of Waves and Particles*, 2nd ed., Springer, 1982.
- [55] F. Pollmann and A. M. Turner, *Detection of symmetry-protected topological phases in one dimension*, Physical Review B 86 (2012), 125441. doi:10.1103/PhysRevB.86.125441.

- [56] M. Prähofer and H. Spohn, *Scale invariance of the PNG droplet and the Airy process*, Journal of Statistical Physics 108 (2002), 1071–1106. doi:10.1023/A:1019791415147.
- [57] A. Pressley and G. Segal, *Loop Groups*, Oxford Mathematical Monographs, Clarendon Press, 1986.
- [58] D. Quillen, *On the formal group laws of unoriented and complex cobordism theory*, Bulletin of the American Mathematical Society 75 (1969), 1293–1298. doi:10.1090/S0002-9904-1969-12401-8.
- [59] M. Reed and B. Simon, *Methods of Modern Mathematical Physics II: Fourier Analysis, Self-Adjointness*, Academic Press, 1975.
- [60] T. Regge and J. A. Wheeler, *Stability of a Schwarzschild singularity*, Physical Review 108 (1957), 1063–1069. doi:10.1103/PhysRev.108.1063.
- [61] E. Seneta, *Non-negative Matrices and Markov Chains*, 2nd ed., Springer Series in Statistics, Springer, 1981.
- [62] J.-P. Serre, *Géométrie algébrique et géométrie analytique*, Annales de l’Institut Fourier 6 (1956), 1–42. doi:10.5802/aif.59.
- [63] V. P. Spiridonov, *On the elliptic beta function*, Russian Mathematical Surveys 56 (2001), 185–186. doi:10.1070/RM2001v056n01ABEH000374.
- [64] M. E. Taylor, *Partial Differential Equations I: Basic Theory*, 2nd ed., Applied Mathematical Sciences 115, Springer, 2011.
- [65] R. Thom, *Quelques propriétés globales des variétés différentiables*, Commentarii Mathematici Helvetici 28 (1954), 17–86. doi:10.1007/BF02566923.
- [66] D. J. Thouless, M. Kohmoto, M. P. Nightingale, and M. den Nijs, *Quantized Hall conductance in a two-dimensional periodic potential*, Physical Review Letters 49 (1982), 405–408. doi:10.1103/PhysRevLett.49.405.
- [67] C. A. Tracy and H. Widom, *On orthogonal and symplectic matrix ensembles*, Communications in Mathematical Physics 177 (1996), 727–754. doi:10.1007/BF02099545.
- [68] V. G. Turaev, *Quantum Invariants of Knots and 3-Manifolds*, de Gruyter Studies in Mathematics 18, de Gruyter, 1994.
- [69] R. M. Wald, *General Relativity*, University of Chicago Press, 1984.
- [70] E. P. Wigner, *On unitary representations of the inhomogeneous Lorentz group*, Annals of Mathematics 40 (1939), 149–204. doi:10.2307/1968551.
- [71] E. Witten, *Elliptic genera and quantum field theory*, Communications in Mathematical Physics 109 (1987), 525–536. doi:10.1007/BF01208956.
- [72] N. M. J. Woodhouse, *Geometric Quantization*, 2nd ed., Clarendon Press, 1992.
- [73] S. L. Woronowicz, *Compact matrix pseudogroups*, Communications in Mathematical Physics 111 (1987), 613–665. doi:10.1007/BF01219077.